LTP02-245-13C
THERMAL PRINTER MECHANISM
TECHNICAL REFERENCE

Rev.05

Seiko Instruments Inc.

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PREFACE

This technical reference (hereinafter referred to as "manual") describes the specifications and basic operating procedures for the LTP02-245-13C thermal printer mechanism (hereinafter referred to as "printer").

"CHAPTER 1 PRECAUTIONS" describes safety, design, and handling precautions. Read it thoroughly before designing so that you can use the product properly.

SII has not investigated the intellectual property rights of the example circuits included in this manual. Fully investigate the intellectual property rights of these circuits before using.

In this manual, the description of negative logic has an additional "/" in the beginning of the terminal name. The input/output direction is described from the product side.

The printer complies with EU RoHS Directive (2011/65/EU).
Also, it contains lead as an exempted application of EU RoHS.

The 10 substances regulated by the EU RoHS Directive

- Cadmium and its compounds
- Chromium (VI) compounds
- Mercury and its compounds
- Lead and its compounds
- Polybrominated biphenyls (PBB)
- Polybrominated diphenyl ethers (PBDE)
- Bis (2-ethylhexyl) phthalate (DEHP)
- Benzyl butyl phthalate (BBP)
- Dibutyl phthalate (DBP)
- Diisobutyl phthalate (DIBP)

The following figure shows parts name of the printer mechanism.

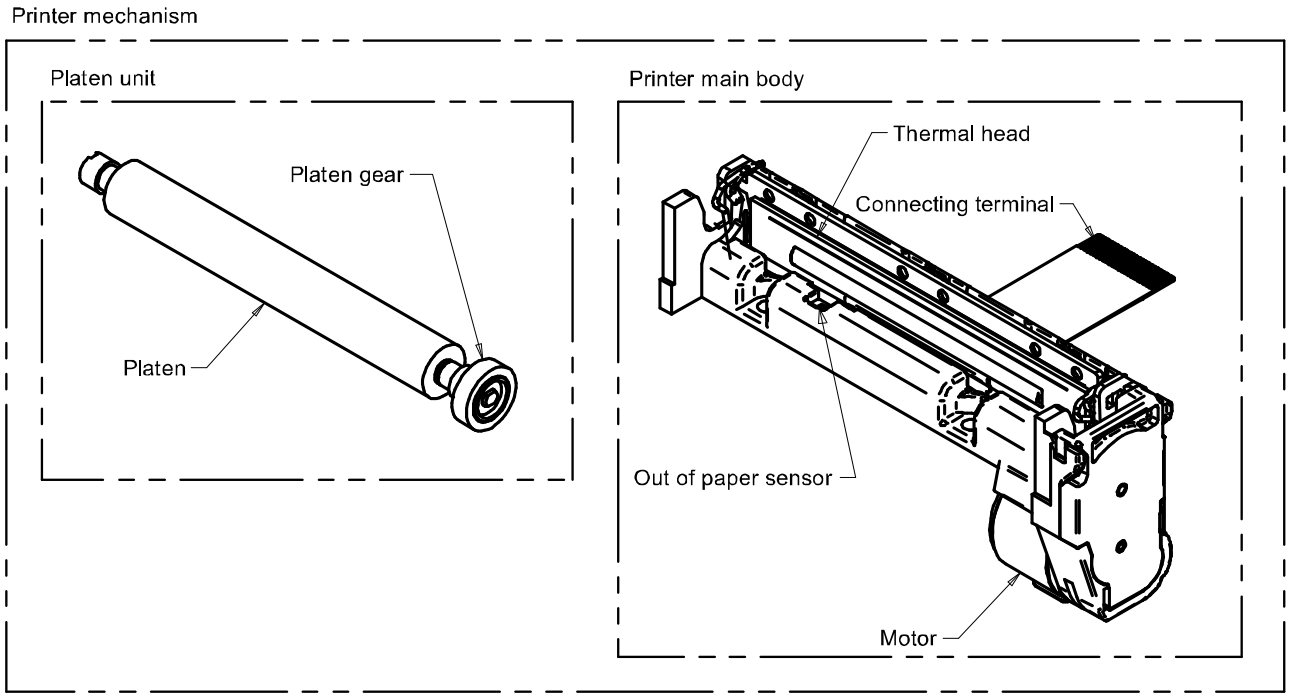


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CHAPTER 1
PRECAUTIONS

Read through this manual to design and to operate the printer properly. Pay special attention to the precautions noted in each section for details. Information contained in this manual is subject to change without notice.
For the latest information, contact our sales representative.

Sufficient evaluation and confirmation should be performed with the designed outer case mounted, to ensure proper use of the printer.

SII shall not be liable for any and all claims, actions, lawsuits, demands, costs, liabilities, losses, damages and/or expenses that are caused by improper handling of the printer, any use not contained in this manual or that result from the outer case, unless such damages and/or loss originate from the printer itself.

SII makes no warrant that your products into which built the example circuits included in this manual can work properly and safe. You shall evaluate and confirm sufficiently that such products can work properly and safe, and shall be liable for any and all claims, actions, lawsuits, demands, costs, liabilities, losses, damages and/or expenses arising out of or in relating to such products.

SII has not investigated the intellectual property rights of the example circuits included in this manual. Fully investigate the intellectual property rights of these circuits before using.

The printer is designed and manufactured to be mounted onto general electronic equipment. When high reliability is required of the printer in respect to hazardous influences on the body or life and loss to property, redundant design of the entire system should be carried out and verify the performance with your actual device before commercialization. And our sales representative should be informed as such in advance.

In order to ensure safe use, design the printer with due consideration of following the precautions. Include any necessary precautions into your operation manual to ensure the safe operation of your product by users.

1.1 SAFETY PRECAUTIONS

In order to ensure safe use, design the printer with due consideration of following the precautions. Include any necessary precautions into your operation manual and attach warning labels to your products to ensure safe operation.

- **Precautions to prevent the thermal head from overheating**

  When the thermal head heat elements are continuously activated by a CPU or other malfunction, the thermal head may overheat and may cause smoke and fire. Be sure to follow "3.5.9 Detecting Abnormal Temperature of Thermal Head" to monitor the temperature of the thermal head to prevent overheating. Turn the printer off immediately when any abnormal conditions occur.

- **Precautions for temperature rise of the thermal head**

  Temperature of the thermal head and its peripherals rises very high during and immediately after printing. Be sure to design the outer case to prevent users from burn injuries by directly touching them and call attention of users by giving sufficient warning with a label or other. Sufficiently warn users for cleaning the thermal head after the thermal head has cooled down. In order to effectively release heat, provide space between the thermal head and the outer case.

- **Precautions for temperature rise of the motor**

  Temperature of the motor and its peripherals rises very high during and immediately after printing. Be sure to design the outer case to prevent users from burn injuries by directly touching them and call attention of users by giving sufficient warning with a label or other. In order to effectively release heat, provide space between the motor and the outer case.

- **Precautions for sharp edges of the printer**

  The printer has some sharp edges and cutting surfaces of the metal parts and those may cause injuries by handling. Be sure to design the outer case to prevent users from touching the sharp edges or call attention of users by giving sufficient warning with labels or others.

- **Precautions for motor drive**

  Your hair may get caught in the exposed platen and the gears. Control the motor not to drive when the outer case and the platen unit is in open state. Also, make sure to design the outer case so that users would not touch the platen and the gears, and as to prevent any objects from getting caught. Call attention of users by giving sufficient warning with a label or others.

1.2 DESIGN AND HANDLING PRECAUTIONS

To maintain the primary performance of the printer and to prevent future problems from occurring, follow the precautions below.

1.2.1 Design Precautions

- Apply power in the following order:
  At power on : (1) V_{dd} → (2) V_p
  At power off : (1) V_p → (2) V_{dd}

- A surge voltage between V_p and GND should not exceed 10 V.

- For noise prevention, put a capacitor of about 0.1 µF in capacitance between V_{dd} and GND near the connector that connects the connecting terminals.

- Make the wiring resistance from V_p and GND to the connecting terminals as small as possible. As a rule of thumb, set the value to 50 mΩ or less. It is recommended to design with keeping a distance from signal lines to reduce interference to the signal system.

- The sensor may generate an instantaneous abnormal signal. Design the firmware in order not to trigger a malfunction caused by the abnormal signal.

- Design the Vp power to become the power off when not printing (including at standby) in order to prevent the thermal head from electrolytic corrosion. In addition, design the signal ground (GND) of the thermal head and the frame ground (FG) of the printer to have the same electric potential.

- Use C-MOS IC chips for CLK, /LAT, DI and DST signals of the thermal head.

- When turning the power on or off, or during not printing, always disable the DST.

- To prevent the thermal head from being damaged by static electricity:
  (1) Connect the printer Frame Ground (FG) connecting area to the Frame Ground (FG) of the outer case. See "CHAPTER 6 OUTER CASE DESIGN GUIDE" for details.
  (2) Connect the Signal Ground (GND) to the Frame Ground (FG) through approximately 1 MΩ resistance.

- Always detect the outputs of the platen position sensor and out-of-paper sensor. Never activate the thermal head when there is no thermal paper. Incorrect activation of the thermal head may significantly reduce the life of the thermal head and the platen and may damage them.

- Secure a pause time of at least 0.5 ms after activating the thermal head and before activating the same heat element. Pay attention especially when performing batch printing or when the thermal head activation time becomes long. When activating continuously without the pause time, the thermal head may become damaged.

- When excessive energy is applied to the thermal head, it may overheat and become damaged. Always use the printer with the specified amount of energy shown in "3.5 CONTROLLING ACTIVATION PULSE WIDTH FOR THERMAL HEAD".

- As the motor drive pulse rate changes, the operating sound and vibration change. Verify the performance with your actual device in advance.

- As the motor drive pulse rate changes, paper feed force may decrease. Verify the performance with your actual device in advance.

- Do not perform continuous printing to prevent the motor from overheating. See "3.3.5 Motor Drive Method" to set a pause time.

- Improper paper feeding may occur in first several dot-lines when printing is started from waiting status or after printing and paper feeding are interrupted. When printing bit images and so on, always feed the thermal paper for more than 48 steps (12 dot-line) at start up and do not interrupt printing.

- To prevent degradation in print quality due to the backlash of the paper feeding drive system, perform printing after feeding the thermal paper for 48 steps (12 dot-line) or more at the initialization, after opening and closing the platen unit, after feeding the thermal paper backward, or after cutting by the paper cutter.
- The surface of the thermal paper may get scratched by backward feed. The backward feed may also cause paper skew and jams depending on the paper roll layout or structure of the paper path. Verify the performance with your actual device in advance.
- When printing at a high print rate for longer length, non-printing area may be colored due to accumulated heat in the thermal head. Verify the performance with your actual device.
- Do not print without the thermal paper. It may cause damage the platen and the thermal head.
- It is recommended that the platen holder is guided by part of the outer case, when the platen unit is set into the printer main body. Setting the mounting part of the printer and rotation center part of the platen holder to the same base, improves accuracy of opening and closing the platen unit.
- Design the outer case and the platen holder to push the center of the platen unit. When the edge of the platen unit is pushed, only one side of the platen unit is locked, it may cause defects such as print failure or paper jam. In order to let users push its center area, any mark indication on the outer case is recommended. Verify the performance with your actual device in advance.
- When the printer main body and the platen unit are not placed in proper position, the print defect and the paper jam may occur. Therefore, pay special attention to it when designing the outer case. For the position relation between the printer main body and the platen unit, see "CHAPTER 6 OUTER CASE DESIGN GUIDE".
- Design the platen holder and the outer case strong enough so that the allowable dimensions can be maintained because stress is applied to them when opening and closing the platen unit. Design the platen holder's axis of rotation not to be rattle as less as possible.
- Depending on the thermal paper feeding position, the surface of the thermal paper may be scratched, or printing failure or abnormal detection of the thermal paper may be caused. Design the outer case in accordance with "Chapter 6 OUTER CASE DESIGN GUIDE". Verify performance with your actual device.
- Label, 2-ply thermal paper, or thermal paper with a thickness of 75 µm or more cannot be used.
- Design the product not to apply any tension force to the FPC. The FPC may be moved by opening and closing the platen unit, so design the product to make the FPC has enough play after connecting. The tension force may cause some print problems and may damage the FPC.
- Metal parts may be discolored and rusted depending on the operational environment. Consider these factors regarding appearance.
- Do not apply external force continuously toward the direction the platen unit is opened by such as a spring. Otherwise, it may cause slipping gears, negative influence on printing quality, and printing troubles during operation.
- When the motor is in a locked state such as excitation, the platen gear may be damaged if the platen unit is blocked from opening and the thermal paper is tried to be forcibly pulled out. Design motor excitation timing and exterior construction with the aforementioned considerations.

1.2.2 Handling Precautions

Incorrect handling may significantly reduce the efficiency of the printer and cause damage. Handle the printer with the following precautions.
Also, include any necessary precautions into your operation manual so that users can handle the printer with care.

- Using anything other than the specified thermal paper does not guarantee print quality and life of the thermal head.
  The followings are examples of trouble:
  (1) Deterioration in print quality due to low thermal sensitivity
  (2) Abrasion of the thermal head due to surface roughness of the thermal paper
  (3) Printing stuck and unusual noise due to sticking the thermal layer of the thermal paper and the thermal paper
  (4) Printing fade due to low preservability the thermal paper
  (5) Electrolytic corrosion of the thermal head due to inferior the thermal paper

- After the printer has been left not in use for long period of time, the platen may be deformed and resulted in print quality deteriorated. In this case, feed thermal paper for a while to recover deformation of the platen. When the thermal head has remained in contact with the platen without thermal paper for a long time, the platen and the thermal head may be stuck together and may not restart operation. When facing this problem, open the platen unit, and close it again.

- Never loosen or unscrew the screws that fasten respective parts of the printer. Loosened screws may significantly reduce the efficiency of the performance of the printer mechanism.

- Do not open the platen unit during printing; otherwise this may significantly reduce the efficiency of the printer and may cause damage in some cases.

- Do not apply stress to the platen unit during printing. The print defect may occur.

- When setting the platen unit, the tips of the platen gear and those of the reduction gear may contact and the platen cannot be installed. In such a case, open the platen unit and close it.

- Never pull out the thermal paper while the platen unit is set. The printer mechanism may be damaged.

- When handling the printer, make sure to use antistatic clothing and to ground yourself to prevent the thermal head from damaged by static electricity. Pay special (particular) attention to the thermal head heat element and the connecting terminals.

- Do not hit or scratch the surface of the thermal head with any sharp or hard object. It may damage the thermal head.

- When printing at a high print rate in a low temperature or high humidity environment, the vapor from the thermal paper during printing may cause condensation on the printer mechanism or soil the thermal paper itself. Pay attention not to fall a drop of water on the thermal head. It may cause electrolytic corrosion of the thermal head. In case of dew condensation, do not turn on the power until the dew dries.

- Always turn off the power of the printer before connecting or disconnecting the connecting terminals.

- Do not apply stress to the FPC when connecting or disconnecting the connecting terminals. Otherwise the FPC may be damaged.

- Call sufficient attention of users to possibility of print defect or paper jam if the thermal paper is pulled or the paper outlet direction is changed during printing and cutting.

- In order to prevent the printer from damage and to avoid the print defect, call sufficient attention of users not to touch the thermal head and the sensor directly when handling the printer such as for replacing the thermal paper.

- Do not use the roll type thermal paper with glued end or folded end. If such the roll type thermal paper needs to be used, replace it with a new roll early before using up the paper.
- The printer is not waterproof or drip proof. Prevent contact with water and do not operate with wet hands. It may cause damage, heating or firing due to short circuit.
- The printer is not dust proof. If the printer is used in a dusty place, the thermal head may be damaged or a paper feeding error may be caused.
- Do not use the printer in corrosive gas and siloxane atmosphere. It may cause the contact failure.

1.2.3 Precautions for Discarding

When discarding used printers, discard them according to the disposal regulations and rules of each respective district.

CHAPTER 2
FEATURES

The printer is a compact size printer mechanism based on the thermal line dot printing method. It can be used for various purpose such as measuring instruments and analyzer, a POS, a communication terminal device, or a data terminal device.

The printer has the following features:

- **High resolution printing**

  A high-density thermal head of 8 dots/mm produces clear and precise printing.

- **Compact**

  Dimensions : W67.3 mm × D18.1 mm × H30.0 mm
  Mass : approx. 28 g

- **High print speed***

  Maximum 100 mm/s print is available.

- **Easy operation**

  Platen unit open mechanism provides easy paper installation.

- **Maintenance Free**

  No cleaning and no maintenance required.

- **Low noise**

  Thermal printing technology realizes low-noise print.

  * Print speed differs depending on working conditions.

CHAPTER 3
SPECIFICATIONS

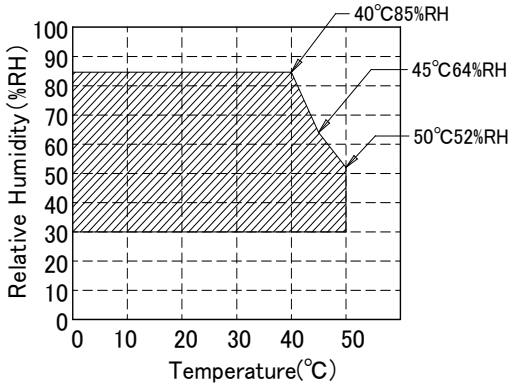
3.1 GENERAL SPECIFICATIONS

Table 3-1 lists the general specifications of the printer.

Table 3-1 General Specifications

(1/2)

Items			Specifications
Printing method			Thermal line dot printing method
Total dots per line			384 dots
Printable dots per line			384 dots
Simultaneously activated dots			45 dots
Resolution			W 8 dots/mm × H 16 dots/mm[*1]
Paper feed pitch			0.03125 mm
Maximum print speed			100 mm/s[*2]
Print width			48 mm
Paper width			58^{0}_{-1} mm
Thermal head temperature detection			Thermistor
Platen position detection			None
Out-of-paper detection			Reflection type photo interrupter
Operating voltage range	V_p line		5.5 to 9.5 V
	V_{dd} line		3.0 to 3.6 V
Current consumption	V_p line	Thermal head drive	2.51 A max. (at 9.5 V)[*3]
		Motor drive	0.60 A max.
	V_{dd} line	Thermal head logic	0.10 A max.

Items		Specifications
Operating temperature range		-10°C to 50°C (Non condensing)
Operating humidity range		
Storage temperature range		-20°C to 60°C (Non condensing)
Life span (at 25°C and rated energy)	Activation pulse resistance	100 million pulses or more*4
	Abrasion resistance	50 km or more*5
Paper feed force		0.49 N (50 gf) or more
Paper hold force		0.78 N (80 gf) or more
Dimensions (excluding convex part)		W 67.3 mm × D 18.1 mm × H 30.0 mm
Mass		Approx. 28 g
Specified thermal paper		Nippon Paper TF50KS-E2D Mitsubishi Hi-Tech Paper P5046

*1 : See "CHAPTER 5 PRINT DRIVE METHOD" for printing drive method.
*2 : Print speed changes according to the processing speed of the controller and print pulse width.
*3 : The value when the number of simultaneously activated dots is 45 dots.
*4 : Excluded when the same dots are printed continuously.
*5 : Excluding damage caused by dust and foreign materials.

3.2 PRINT CONFIGURATION

Figure 3-1 shows dot pitch. Figure 3-2 shows print area.

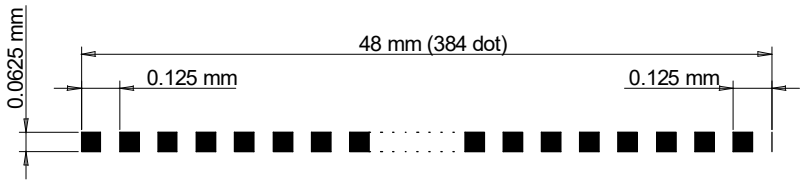


Figure 3-1 Dot Pitch

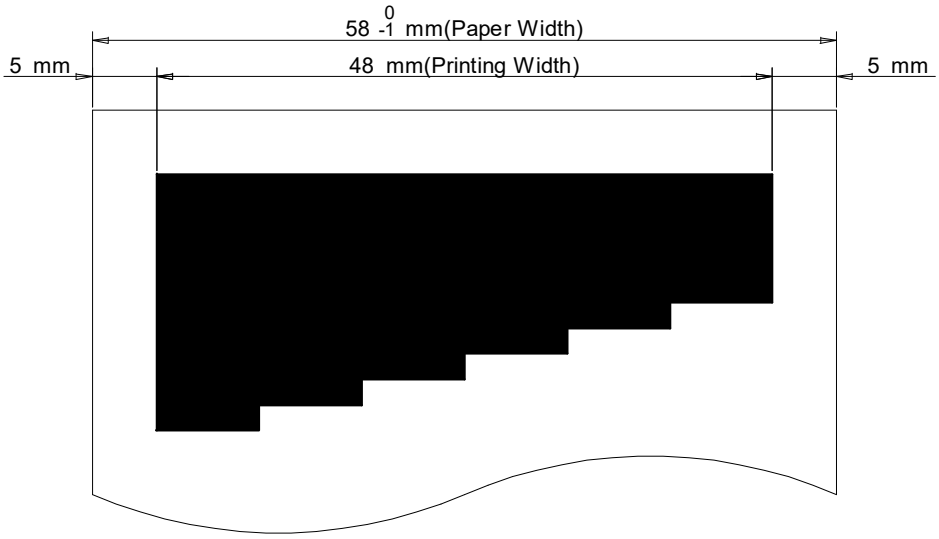


Figure 3-2 Print Area

3.3 STEP MOTOR

3.3.1 General Specifications

Table 3-2 shows the general specifications of the step motor.

Table 3-2 General Specifications of Step Motor

Item	Specifications
Type	PM type step motor
Drive method	Bipolar chopper drive
Excitation	1-2 phase
Winding resistance per phase	8.3 Ω/phase ±10%
Motor drive voltage	V_p: 5.5 to 9.5 V
Motor controlled current	300 mA/phase
Drive pulse rate	3200 pps max.

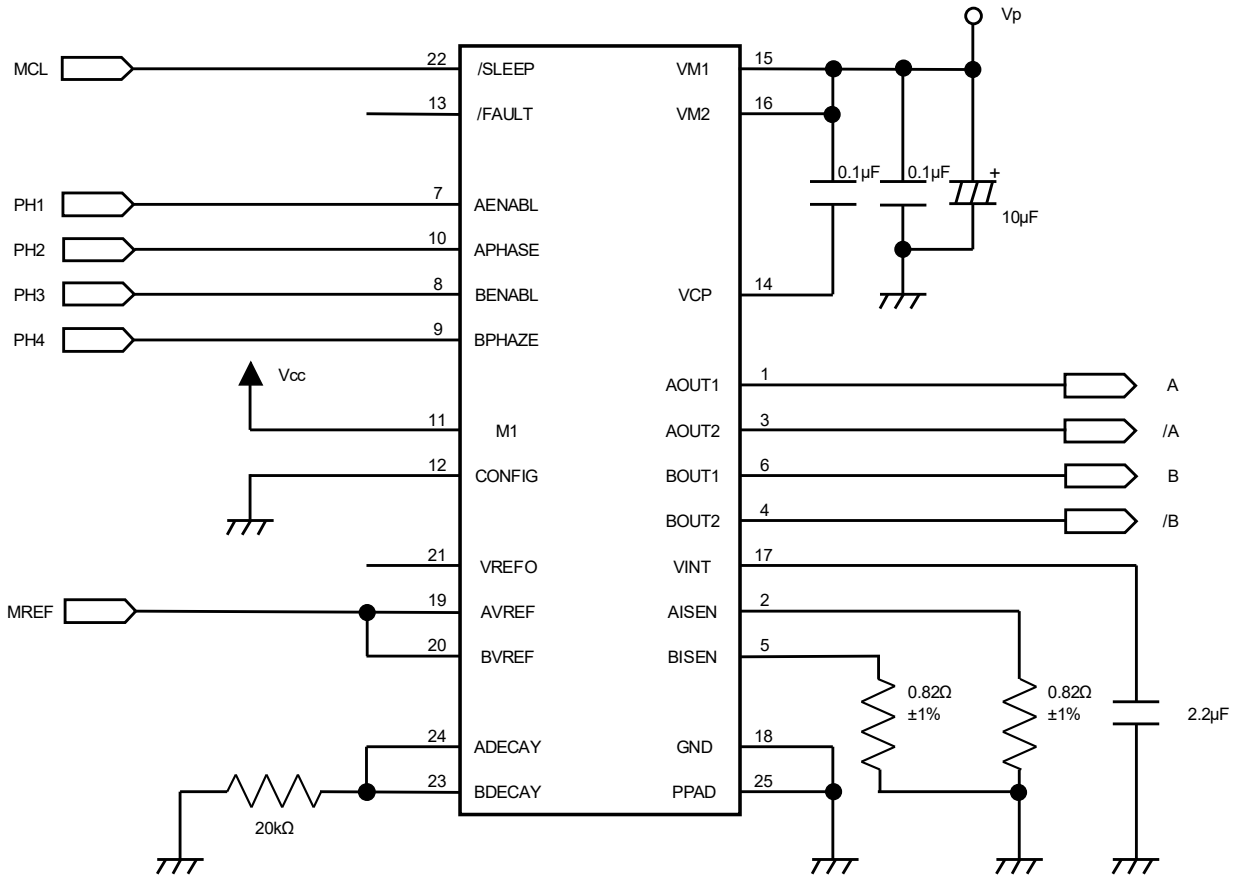
3.3.2 Example Drive Circuit

Figure 3-3 shows an example drive circuit.

- MREF signal

  The MREF signal is a reference signal for the motor current control. The voltage set by the MREF signal determines the drive setting current of the motor.

Motor Drive Setting Current	MREF Signal Voltage
300 mA	1.23 V ±2%

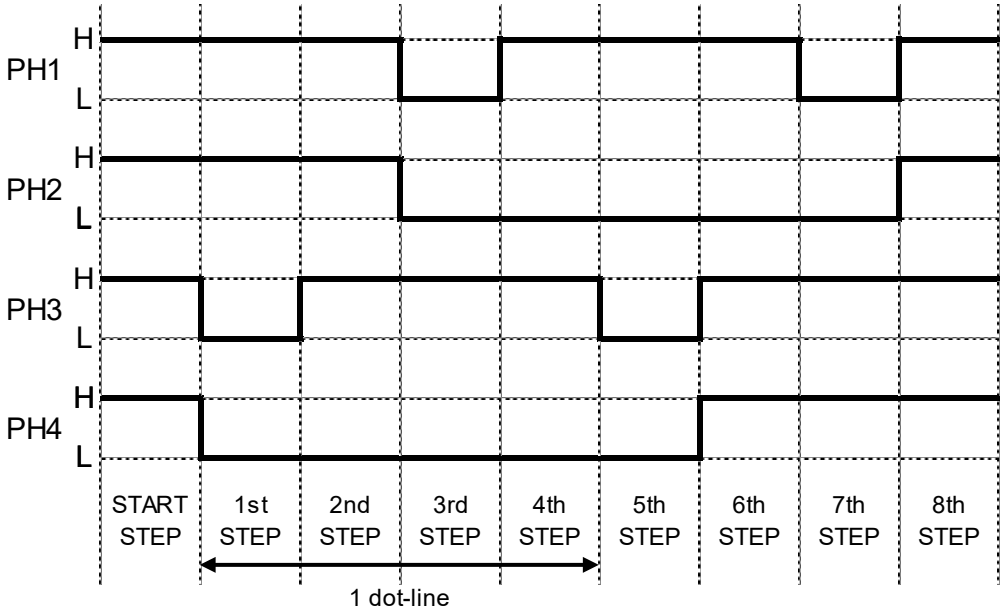


*Recommended motor driver : DRV8834RGER

Figure 3-3 Example Drive Circuit

3.3.3 Excitation Sequence

Drive the motor with 1-2 phase excitation. 1 step of the motor drive signal feeds the paper 0.03125 mm.
1 dot-line consists of 4 steps. When the example drive circuit shown in Figure 3-3 is input the voltage waveforms shown in Figure 3-4, the motor is excited as shown in Table 3-3 in order of step 1, 2, 3, 4, 5, 6, 7, 8, 1, 2, ... , and the printer feeds the paper in the forward direction.



*: Set MCL to "High" while the motor is driven.

Figure 3-4 Input Voltage Waveforms for Example Drive Circuit

Table 3-3 Excitation Sequence

	Input Signal				Output Signal			
	PH1	**PH2**	**PH3**	**PH4**	**A**	**/A**	**B**	**/B**
Step 1	H	H	L	L	H	L	OFF	OFF
Step 2	H	H	H	L	H	L	L	H
Step 3	L	L	H	L	OFF	OFF	L	H
Step 4	H	L	H	L	L	H	L	H
Step 5	H	L	L	L	L	H	OFF	OFF
Step 6	H	L	H	H	L	H	H	L
Step 7	L	L	H	H	OFF	OFF	H	L
Step 8	H	H	H	H	H	L	H	L

3.3.4 Motor Start/Stop Method

Design the control circuit and the software for starting and stopping the motor referring to the timing chart in Figure 3-5. Also note the following precautions:

(1) Start step

To start up the motor from the pause (no excitation) state, shift the motor to the sequence of the print step after exciting the same phase as that of the stop step with the time for the 1st acceleration step time.

To restart the motor from the stop step, immediately shift the motor to the sequence of print step.

(2) Stop step

To stop the motor, excite the same phase as the last one in the printing step for 65 ms.

(3) Pause state

In the pause state, do not excite the motor and prevent the motor from overheating. Even when the motor is not excited, holding torque of the motor prevents the thermal paper from moving.

Figure 3-5 shows input waveforms with the example drive circuit.

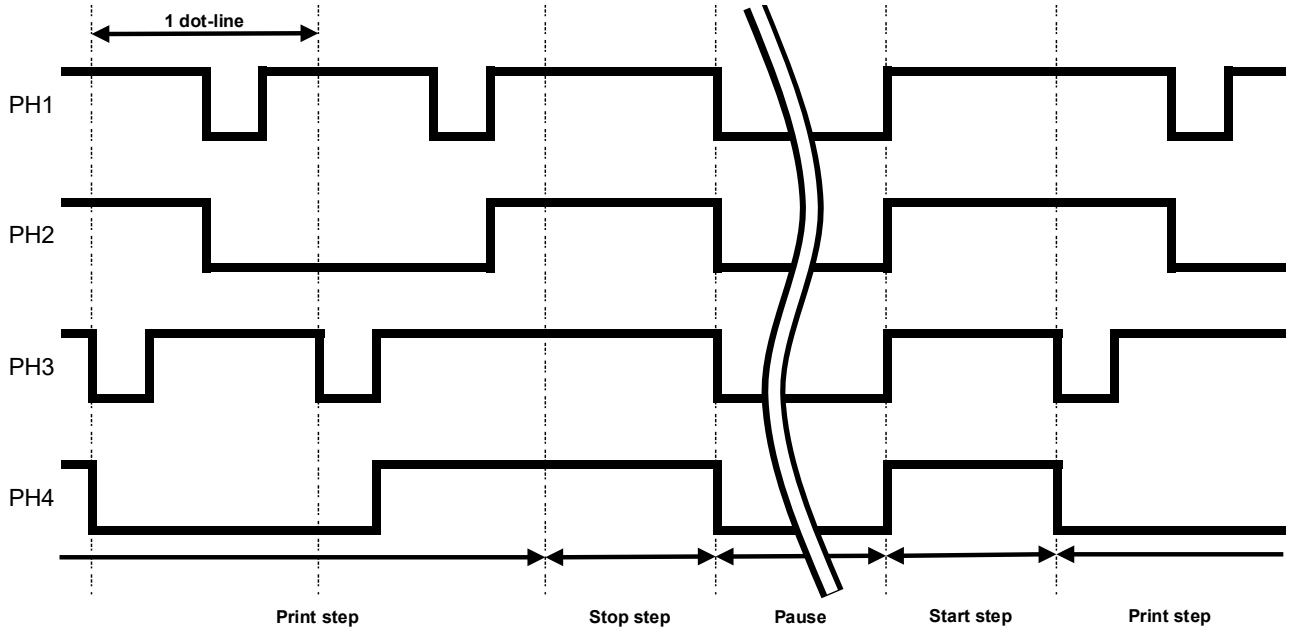


Figure 3-5 Motor Start/Stop Timing Chart

3.3.5 Motor Drive Method

To drive the motor, follow the method below.

(1) Motor drive pulse rate

During paper feeding, drive the motor with less than the maximum motor drive pulse rate P_M to be obtained by equation (1).

Equation (1):

$$P_M = V_p \times 534 - 1339(\text{pps})$$

P_M : Maximum motor drive pulse rate at Vp (pps)
However, 3200 pps max.

V_p : Motor drive voltage (V)

Table 3-4 Maximum Motor Drive Pulse Rate

V_p	Maximum Motor Drive Pulse Rate
5.5 V	1598 pps
6.5 V	2132 pps
7.5 V	2666 pps
8.5 V	3200 pps
9.5 V	3200 pps

During paper feeding in backward, the motor should be driven lower than 960 pps.

(2) Motor speed control

When driving the motor, the acceleration control is required to maintain the paper feed force of start up. When the acceleration is not performed correctly, the motor may not be able to rotate normally when it has a heavy workload. Accelerate the speed sequentially up to the maximum motor drive pulse rate P_M according to the acceleration steps shown in Table 3-5.

Acceleration is performed by exciting the phases with the following acceleration step times.

1. Drive the start step at the acceleration step time of acceleration step 1.
2. Drive the 1st step at the acceleration step time of acceleration step 1.
3. Drive the 2nd step at the acceleration step time of acceleration step 2.
4. Hereinafter, drive the "n"th step at the acceleration step time of the acceleration step "n".
5. After accelerating up to the maximum motor drive pulse rate P_M, drive the motor at a constant speed.

Printing is possible during acceleration.

Follow the procedures below when the acceleration of speed is unable by following the acceleration steps shown in Table 3-5 or when the speed has been reduced at a certain speed and then accelerates the speed again.

The next step time after reducing the speed is the nearest acceleration step time that is longer than the previous acceleration step time and is the step time in the minimum.

(e.g.)
In the case of Table 3-5, when the previous step time is 950 μs, the next step should be the 11th acceleration step (967 μs).

Hereinafter, accelerate the speed sequentially up to the maximum motor drive pulse rate P_M according to Table 3-5.

Table 3-5 Acceleration Steps

Acceleration Steps	Drive Pulse Rate (pps)	Acceleration Step Time (μs)	Acceleration Steps	Drive Pulse Rate (pps)	Acceleration Step Time (μs)
Start	-	4291	49	2270	440
1	233	4291	50	2294	436
2	377	2652	51	2317	432
3	488	2048	52	2341	427
4	582	1719	53	2363	423
5	663	1507	54	2386	419
6	737	1357	55	2409	415
7	805	1243	56	2431	411
8	867	1153	57	2453	408
9	926	1080	58	2475	404
10	981	1019	59	2497	400
11	1034	967	60	2519	397
12	1084	923	61	2540	394
13	1132	883	62	2561	390
14	1178	849	63	2582	387
15	1222	818	64	2603	384
16	1265	790	65	2624	381
17	1307	765	66	2644	378
18	1347	742	67	2665	375
19	1386	721	68	2685	372
20	1424	702	69	2705	370
21	1462	684	70	2725	367
22	1498	668	71	2745	364
23	1533	652	72	2764	362
24	1568	638	73	2784	359
25	1602	624	74	2803	357
26	1635	612	75	2822	354
27	1668	600	76	2842	352
28	1700	588	77	2860	350
29	1731	578	78	2879	347
30	1762	568	79	2898	345
31	1792	558	80	2917	343
32	1822	549	81	2935	341
33	1851	540	82	2954	339
34	1880	532	83	2972	336
35	1909	524	84	2990	334
36	1937	516	85	3008	332
37	1964	509	86	3026	330
38	1992	502	87	3044	329
39	2018	495	88	3062	327
40	2045	489	89	3079	325
41	2071	483	90	3097	323
42	2097	477	91	3114	321
43	2123	471	92	3132	319
44	2148	466	93	3149	318
45	2173	460	94	3166	316
46	2198	455	95	3183	314
47	2222	450	96	3200	313
48	2246	445	-	-	-

(3) Preventing Overheat

To prevent the motor from overheating, the maximum continuous drive time and drive rate are limited. Table 3-6 shows the Maximum Continuous Drive Time and Drive Rate of the motor per print within 1 m. Follow Table 3-6 shown below to set drive time and the pause time of the motor.
Table 3-7 and Table 3-8 show Drive Time and Paper Length at Temperature Rise of 50°C and 75°C.

The temperature rise of the motor varies depending on the operational environment. (ambient temperature, designing the outer case etc.) Keep the temperature of the motor outer case, 100°C or lower. However, do not mount heat sink to the motor. Verify the performance with your actual device.

Table 3-6 Maximum Continuous Drive Time and Drive Rate

Drive Pulse Rate (pps)	Maximum Continuous Drive Time (sec)	Motor Drive Voltage V_p (V)				
		9.5V≥V_p>8.5V	8.5V≥V_p>7.5V	7.5V≥V_p>6.5V	6.5V≥V_p>5.5V	5.5V
		Drive Rate				
320 to 480	100	45%	45%	50%	55%	60%
480 to 640	67	45%	45%	50%	55%	60%
640 to 800	50	45%	45%	50%	55%	60%
800 to 960	40	45%	45%	50%	55%	60%
960 to 1120	33	45%	45%	50%	55%	60%
1120 to 1280	29	45%	45%	50%	55%	60%
1280 to 1440	25	45%	45%	50%	55%	60%
1440 to 1600	22	45%	45%	50%	55%	60%
1600 to 1760	20	45%	45%	50%	55%	-
1760 to 1920	18	45%	45%	50%	55%	-
1920 to 2080	17	45%	45%	50%	55%	-
2080 to 2240	15	45%	45%	50%	55%	-
2240 to 2400	14	45%	45%	50%	-	-
2400 to 2560	13	45%	45%	50%	-	-
2560 to 2720	13	45%	45%	50%	-	-
2720 to 2880	12	45%	45%	-	-	-
2880 to 3040	11	45%	45%	-	-	-
3040 to 3200	11	45%	50%	-	-	-
3200	10	45%	50%	-	-	-

-: Unusable

Set the Drive Pulse Rate according to "3.3.5 Motor Drive Method".

$$\text{Drive Rate (\%)} = \frac{\text{Drive Time}}{\text{Drive Time} + \text{Pause Time}} \times 100\ (\%)$$

Table 3-7 Drive Time and Paper Length at Temperature Rise 50°C

Drive Pulse Rate (pps)	Motor Drive Voltage V_p (V)									
	9.5V≥V_p>8.5V		8.5V≥V_p>7.5V		7.5V≥V_p>6.5V		6.5V≥V_p>5.5V		5.5V	
	min	(m or longer)	min	(m or longer)	min	(m or longer)	min	(m or longer)	min	(m or longer)
320 to 480	5.1	3.1	5.2	3.1	5.1	3.1	4.7	2.8	4.9	2.9
480 to 640	5.1	4.6	5.2	4.7	5.1	4.6	4.8	4.3	5.0	4.5
640 to 800	5.0	6.0	5.1	6.2	5.1	6.1	4.8	5.8	5.2	6.2
800 to 960	4.9	7.4	5.1	7.7	5.1	7.7	4.9	7.3	5.3	8.0
960 to 1120	4.8	8.7	5.1	9.1	5.1	9.2	4.9	8.9	5.4	9.8
1120 to 1280	4.8	10.0	5.0	10.5	5.1	10.8	5.0	10.5	5.6	11.7
1280 to 1440	4.7	11.3	5.0	12.0	5.1	12.3	5.1	12.1	5.7	13.7
1440 to 1600	4.6	12.5	4.9	13.3	5.1	13.9	5.1	13.8	5.8	15.7
1600 to 1760	4.6	13.7	4.9	14.7	5.1	15.4	5.2	15.5	-	-
1760 to 1920	4.5	14.8	4.9	16.0	5.2	17.0	5.2	17.2	-	-
1920 to 2080	4.4	15.9	4.8	17.4	5.2	18.6	5.3	18.9	-	-
2080 to 2240	4.3	16.9	4.8	18.6	5.2	20.1	5.3	20.7	-	-
2240 to 2400	4.3	17.9	4.7	19.9	5.2	21.7	-	-	-	-
2400 to 2560	4.2	18.8	4.7	21.1	5.2	23.3	-	-	-	-
2560 to 2720	4.1	19.7	4.7	22.4	5.2	24.8	-	-	-	-
2720 to 2880	4.0	20.6	4.6	23.6	-	-	-	-	-	-
2880 to 3040	4.0	21.4	4.6	24.7	-	-	-	-	-	-
3040 to 3200	3.9	22.2	4.5	25.9	-	-	-	-	-	-
3200	3.8	22.9	4.5	27.0	-	-	-	-	-	-
-: Unusable										

Set the Drive Pulse Rate according to "3.3.5 Motor Drive Method".

Table 3-8 Drive Time and Paper Length at Temperature Rise 75°C

Drive Pulse Rate (pps)	Motor Drive Voltage V_p (V)									
	9.5V≥V_p>8.5V		8.5V≥V_p>7.5V		7.5V≥V_p>6.5V		6.5V≥V_p>5.5V		5.5V	
	min	(m or longer)	min	(m or longer)	min	(m or longer)	min	(m or longer)	min	(m or longer)
320 to 480	12.1	7.3	12.9	7.8	12.1	7.3	9.2	5.5	9.9	6.0
480 to 640	11.9	10.7	12.9	11.6	12.4	11.2	10.0	9.0	11.3	10.2
640 to 800	11.7	14.0	12.9	15.5	12.7	15.2	10.7	12.9	12.7	15.3
800 to 960	11.5	17.3	12.9	19.3	13.0	19.4	11.5	17.2	14.1	21.2
960 to 1120	11.3	20.3	12.8	23.1	13.2	23.8	12.2	22.0	15.5	28.0
1120 to 1280	11.1	23.3	12.8	26.9	13.5	28.4	13.0	27.2	16.9	35.6
1280 to 1440	10.9	26.2	12.8	30.7	13.8	33.2	13.7	32.9	18.4	44.0
1440 to 1600	10.7	28.9	12.8	34.5	14.1	38.1	14.4	39.0	19.8	53.3
1600 to 1760	10.5	31.5	12.8	38.3	14.4	43.2	15.2	45.6	-	-
1760 to 1920	10.3	34.0	12.8	42.1	14.7	48.4	15.9	52.6	-	-
1920 to 2080	10.1	36.3	12.7	45.9	15.0	53.8	16.7	60.0	-	-
2080 to 2240	9.9	38.6	12.7	49.6	15.2	59.4	17.4	67.9	-	-
2240 to 2400	9.7	40.7	12.7	53.3	15.5	65.2	-	-	-	-
2400 to 2560	9.5	42.7	12.7	57.1	15.8	71.1	-	-	-	-
2560 to 2720	9.3	44.6	12.7	60.8	16.1	77.2	-	-	-	-
2720 to 2880	9.1	46.3	12.6	64.5	-	-	-	-	-	-
2880 to 3040	8.9	47.9	12.6	68.2	-	-	-	-	-	-
3040 to 3200	8.7	49.4	12.6	71.9	-	-	-	-	-	-
3200	8.5	50.8	12.6	75.6	-	-	-	-	-	-
-: Unusable										

Set the Drive Pulse Rate according to "3.3.5 Motor Drive Method".

3.3.6 Precautions for Motor Drive

- Use of a motor drive circuit other than the one shown in "3.3.2 Example Drive Circuit", may not ensure the specified performance.
- To prevent degradation in print quality due to the backlash of the paper feeding drive system, feed the thermal paper for 48 steps (12 dot-line) or more at the initialization after opening and closing the platen unit, after feeding the thermal paper backward, or after cutting with a paper cutter. During this time, drive the motor with constant speed at the 1st acceleration step.
- When printing, change the motor drive pulse rate depending on the operational conditions such as voltage, temperature, and the number of activated dots. (See "CHAPTER 5 PRINT DRIVE METHOD" for details.)
- When printing, change the motor drive pulse rate to make the activation pulse width of the thermal head not to exceed the sum of the 2-step times of the motor. (See "CHAPTER 5 PRINT DRIVE METHOD" for details.)
- Do not perform continuous printing to prevent the motor from overheating. Refer to "3.3.5 Motor Drive Method" to set a pause time.
- The surface of the thermal paper may get scratched by backward feed. The backward feed may cause paper skew and jams depending on the thermal paper layout or structure of the paper path. Verify the performance with your product device in advance.
- Do not print intermittently. (Do not repeat printing and stopping in a short interval.)
  Print quality may be decreased due to unevenness of the paper feed pitch.
- Always perform the start and the stop steps for both character print and bit image print.
- For the motor stop, a minimum 4 steps (1 dot-line) of motor feed is required from the step that thermal head was activated. When the motor is stopped at the step that the thermal head has been activated, it may cause the thermal paper to stick on the thermal head and paper feed difficulty may occur.
- Sound and vibration during printing vary depending on the motor drive pulse rate. Verify the performance with your actual device in advance.

3.4 THERMAL HEAD

The thermal head consists of heat elements and a thermal head driver that drives and controls the heat elements. For the print data entered through the DI, print is "High" and non print is "low".
The data from the DI is transferred to the shift register at the rising edge of the CLK signal.
The data is stored into the latch register by making /LAT signal "Low" after 1 line data is transferred. The heat elements are activated by making DST signal "High" in accordance with the stored print data.

Simultaneously activated dots is 45 dots.

3.4.1 Structure of Thermal Head

Figure 3-6 shows the thermal head block diagram when driving the printer.

Table 3-9 shows the relationship between DST No. and activated heat elements No..

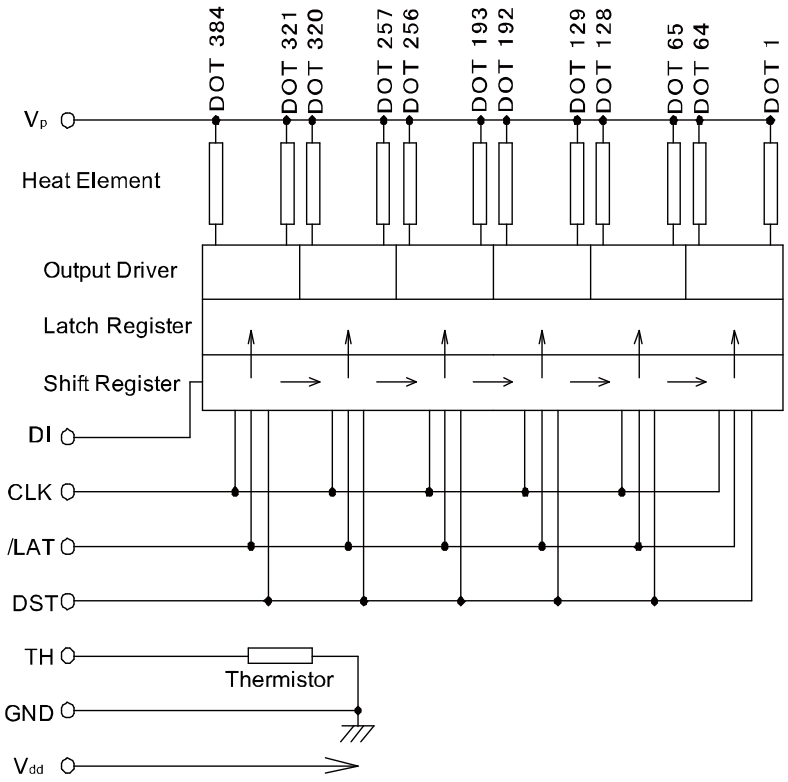


Figure 3-6 Thermal Head Block Diagram

Table 3-9 DST No. and Heat Elements No.

Block	DST No.	Heat Elements No.	Dots/DST
1	DST	1 to 64	384
2		65 to 128	
3		129 to 192	
4		193 to 256	
5		257 to 320	
6		321 to 384	

3.4.2 Relation Between Transfer Data and Print Position

The 384-bit data (#1 to #384) transferred through DI are printed when driving the printer as shown in Figure 3-7.

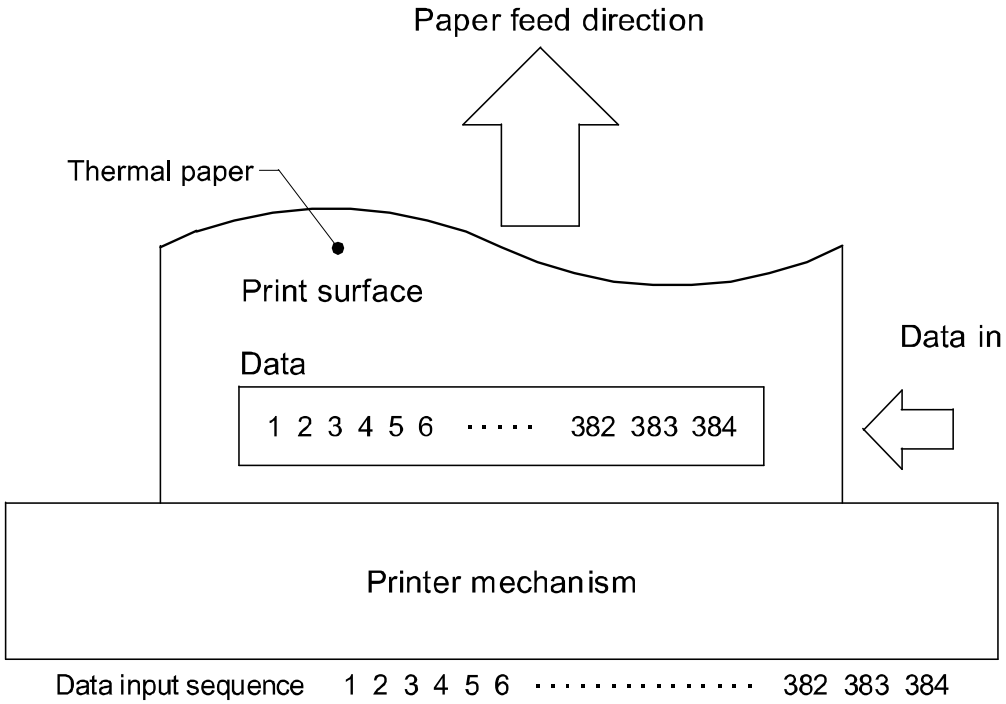


Figure 3-7 Transfer Data and Print Position

3.4.3 Electrical Characteristics of Thermal Head

Table 3-10 shows the electrical characteristics of thermal head.

Table 3-10 Electrical Characteristics of Thermal Head

(Temperature : 25°C)

Item		Symbol	Conditions	Rated Value			Unit
				Min.	Typ.	Max.	
Thermal head heat element resistance		R_H		170.7	176.0	181.3	Ω
Thermal head drive voltage		V_P		5.5	-	9.5	V
Thermal head drive current		I_P	The number of simultaneously activated dots = 45	-	-	2.51	A
Logic voltage		V_{dd}		3.0	3.3	3.6	V
Logic current		I_{dd}	f_{DI} = 1/2f_{CLK}, f_{CLK} = 8 MHz	-	-	20	mA
Input voltage	High	V_{IH}	CLK, DI, /LAT, DST	0.8 × V_{dd}	-	V_{dd}	V
	Low	V_{IL}	CLK, DI, /LAT, DST	0	-	0.2 × V_{dd}	V
DI input current	High	I_{IH} DI	V_{IH} = V_{dd}	-	-	0.5	μA
	Low	I_{IL} DI	V_{IL} = 0 V	-	-	-0.5	μA
DST input current (High active)	High	I_{IH} DST	V_{IH} = V_{dd}	-	-	330.0	μA
	Low	I_{IL} DST	V_{IL} = 0 V	-	-	-3.0	μA
CLK input current	High	I_{IH} CLK	V_{IH} = V_{dd}	-	-	1.0	μA
	Low	I_{IL} CLK	V_{IL} = 0 V	-	-	-1.0	μA
/LAT input current	High	I_{IH} /LAT	V_{IH} = V_{dd}	-	-	1.0	μA
	Low	I_{IL} /LAT	V_{IL} = 0 V	-	-	-110.0	μA
CLK frequency		f_{CLK}	-	-	-	10	MHz
CLK pulse width		t1[*1]	-	40	-	-	ns
DI setup time		t2[*1]	-	40	-	-	ns
DI hold time		t3[*1]	-	40	-	-	ns
/LAT setup time		t4[*1]	-	100	-	-	ns
/LAT pulse width		t5[*1]	-	100	-	-	ns
/LAT hold time		t6[*1]	-	100	-	-	ns
DST setup time		t7[*1]	-	100	-	-	ns
/LAT wait time		t8[*1,*2]	-	15	-	-	μs

*1: See the timing chart.
*2: When MIN at "/LAT wait time" in the table cannot be secured, it may cause V_p voltage fluctuations.

3.4.4 Timing Chart

Figure 3-8 shows the thermal head timing chart of the printer.

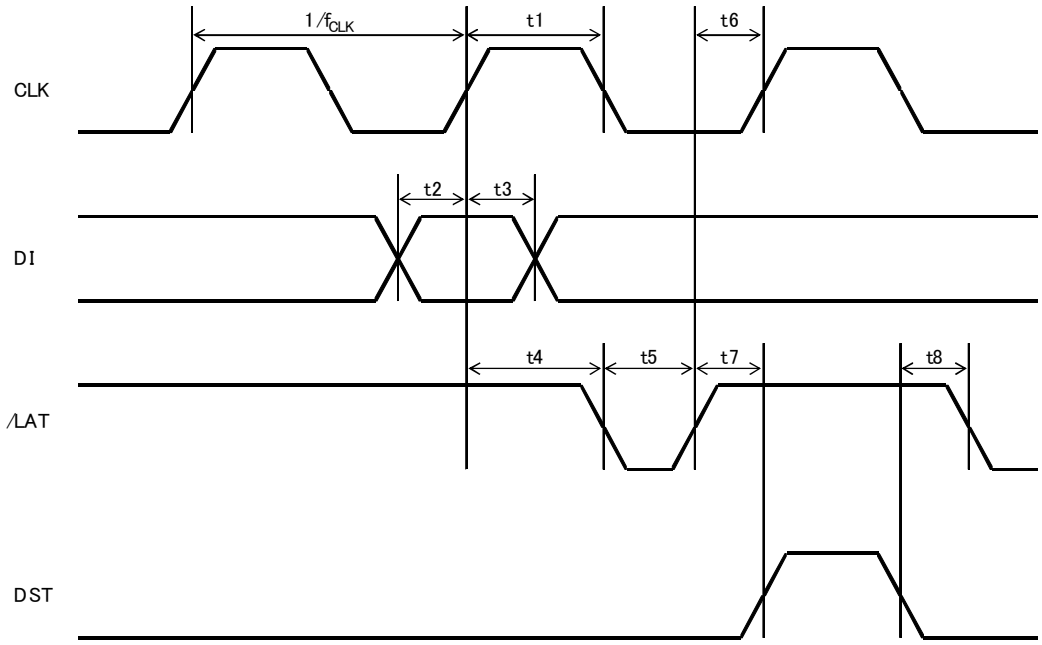


Figure 3-8 Thermal Head Timing Chart

3.4.5 Thermal Head Heat Element Resistance

Table 3-11 shows the thermal head heat element resistance value of the printer.

Table 3-11 Thermal Head Heat Element Resistance

Thermal Head Heat Element Resistance
170.7 to 181.3 Ω

3.4.6 Maximum Current Consumption

Since the maximum current consumption during the thermal head drive reaches the value of the equation (2) when the thermal head is driven, the number of simultaneously activated dots should be determined not to exceed power supply capacity. Also, allowable current for the cable material and the voltage drop on the cable should be cared well.

Equation (2):

$$I_p = \frac{N_{SA} \times V_p}{R_{Hmin}}$$

Ip	:	Maximum current consumption (A)	
N_{SA}	:	Number of simultaneously activated dots	
V_p	:	Thermal head drive voltage (V)	
R_{Hmin}	:	Minimum thermal head heat element resistance	170.7 (Ω)

3.5 CONTROLLING ACTIVATION PULSE WIDTH FOR THERMAL HEAD

To execute high quality printing by using the printer, the activation pulse width is controlled according to printer use conditions. Control printing with the activation pulse width calculated by the following sequence. Printing at too high voltage or too long activation pulse width may significantly shorten the life of the thermal head.

3.5.1 Calculation of Activation Pulse Width

The activation pulse width "t" can be calculated by substituting each calculated value according to the procedures shown in "3.5.2 Setting of Printing Energy" to "3.5.6 Adjustment by Thermal Head Activation Pulse Cycle" in the equation (3).

Equation (3):

$$t = \frac{E \times R}{V^2} \times C$$

t	:	Activation pulse width (ms)	
E	:	Printing energy (mJ)	See section 3.5.2
R	:	Adjustment resistance (Ω)	See section 3.5.3
V	:	Adjustment voltage (V)	See section 3.5.4
C	:	Adjustment coefficient by activation pulse cycle	See section 3.5.6

3.5.2 Setting of Printing Energy

Calculate the printing energy "E" using the equation (4) as the appropriate printing energy differs depending on each specified thermal paper and the temperature of the thermal head.

Equation (4):

$$E = E_{25} - T_C \times (T_X - 25)$$

E_{25}	:	Standard printing energy (mJ)	See Table 3-12
T_C	:	Temperature coefficient	See Table 3-12
T_X	:	Temperature detected by thermistor (°C)*	

*: Measure the temperature using the resistance of the built-in thermistor on the thermal head.
For the thermistor resistance value at T_X (°C), see "3.5.8 Temperature Characteristics of Thermistor".

Table 3-12 Standard Printing Energy and Temperature Coefficient

Thermal paper		Standard Printing Energy (mJ)	Temperature Coefficient
Nippon Paper	TF50KS-E2D	0.2867	0.003430
Mitsubishi Hi-Tech Paper	P5046	0.3520	0.002777

3.5.3 Adjustment of Thermal Head Resistance

Calculate the adjusted resistance "R" using equation (5) as a voltage drop is caused by wiring resistance.

Equation (5):

$$R = \frac{(R_H + R_i + (R_c + r_c) \times N_{SA})^2}{R_H}$$

R_H	:	Thermal head heat element resistance	180.0 (Ω)
R_i	:	Wiring resistance in the thermal head	9.0 (Ω)
R_C	:	Common terminal wiring resistance in the thermal head	0.278 (Ω)
r_C	:	Wiring resistance between Vp and GND (Ω)*	
N_{SA}	:	The number of simultaneously activated dots	

*: The resistance is a serial resistance of the wire and switching circuit of relay between control terminal and power supply.

3.5.4 Adjustment of Thermal Head Applied Voltage

Calculate the adjustment applied voltage "V" using equation (6) as printing density is changed by the difference of applied voltage to the thermal head.

Equation (6):

$$V = 1.124 \times V_p - 1.182$$

V_p : Thermal head drive voltage (V)

3.5.5 Setting of Activation Pause Time

In order to protect the thermal head heat elements, when the same heat element dots are activated continuously on the successive dot-line, determine the activation pulse cycle (the time from the start of the preceding activation to the start of the current activation) which meets equation (7) to secure the pause time.

Equation (7):

$$W > t + 500\ (\mu s)$$

W : The activation cycle (μs)*

*: The activation cycle W is the driving time of the printer drive motor for 2 steps (1-half dot-line).

3.5.6 Adjustment by Thermal Head Activation Pulse Cycle

Calculate the adjustment coefficient "C" by thermal head activation pulse cycle using equation (8) as print density is changed by the difference of the thermal head activation pulse cycle (equivalent for motor drive pulse rate).

Equation (8):

$$C = 1 - \frac{2230 \times V_p - 6429}{(V_p - 3.78) \times (W' + 2420) + 2100}$$

W' : The time between the beginning of the last thermal head activation and the beginning of present thermal head activation (μs)

To calculate the activation pulse width t1 for the half dot-line 2 (at step 3 and 4), W'1 shown in the figure below should be set as W' defined at Equation (8).

To calculate the activation pulse width t2 for the half dot-line 5 (at step 9 and 10), W'2 shown in the figure below should be set as W' defined at Equation (8).

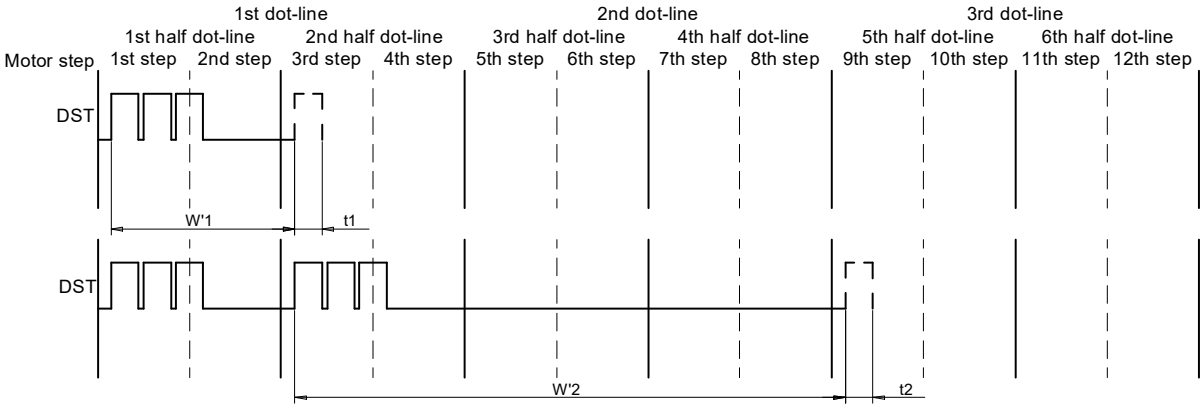


Figure 3-9 Time Between Beginning of Last Thermal Head Activation and Beginning of Present Thermal Head Activation (W')

3.5.7 Calculation Example for Activation Pulse Width

Table 3-13 lists the calculation examples of the activation pulse width calculated using equation (3) and the values obtained using equations (4) to (6) and (8).

Table 3-13 Activation Pulse Width

Unit : ms

V_p [V]	T_X [°C]	Motor Drive Pulse Rate [pps]								
		640	960	1280	1600	1920	2240	2560	2880	3200
5.5	-10	1.830	1.495	1.277	-	-	-	-	-	-
	0	1.676	1.369	1.169	-	-	-	-	-	-
	10	1.521	1.243	1.062	-	-	-	-	-	-
	20	1.367	1.117	0.954	-	-	-	-	-	-
	30	1.213	0.991	0.846	-	-	-	-	-	-
	40	1.058	0.865	0.739	-	-	-	-	-	-
	50	0.904	0.739	0.631	-	-	-	-	-	-
	60	0.750	0.612	0.523	-	-	-	-	-	-
	70	0.595	0.486	0.416	-	-	-	-	-	-
6.5	-10	1.298	1.071	0.921	0.813	0.733	-	-	-	-
	0	1.189	0.981	0.843	0.745	0.671	-	-	-	-
	10	1.079	0.891	0.765	0.676	0.609	-	-	-	-
	20	0.970	0.800	0.688	0.608	0.548	-	-	-	-
	30	0.860	0.710	0.610	0.539	0.486	-	-	-	-
	40	0.751	0.620	0.533	0.470	0.424	-	-	-	-
	50	0.641	0.529	0.455	0.402	0.362	-	-	-	-
	60	0.532	0.439	0.377	0.333	0.300	-	-	-	-
	70	0.422	0.349	0.300	0.265	0.239	-	-	-	-
7.5	-10	0.955	0.793	0.684	0.605	0.546	0.500	0.463	-	-
	0	0.875	0.726	0.626	0.554	0.500	0.458	0.424	-	-
	10	0.794	0.659	0.568	0.503	0.454	0.416	0.385	-	-
	20	0.714	0.592	0.511	0.452	0.408	0.373	0.346	-	-
	30	0.633	0.525	0.453	0.401	0.362	0.331	0.307	-	-
	40	0.553	0.459	0.395	0.350	0.316	0.289	0.268	-	-
	50	0.472	0.392	0.338	0.299	0.270	0.247	0.229	-	-
	60	0.391	0.325	0.280	0.248	0.224	0.205	0.190	-	-
	70	0.311	0.258	0.222	0.197	0.178	0.163	0.151	-	-
8.5	-10	0.729	0.607	0.524	0.465	0.420	0.385	0.356	0.333	0.314
	0	0.668	0.556	0.480	0.426	0.384	0.352	0.326	0.305	0.287
	10	0.606	0.505	0.436	0.386	0.349	0.320	0.296	0.277	0.261
	20	0.545	0.453	0.392	0.347	0.314	0.287	0.266	0.249	0.234
	30	0.483	0.402	0.348	0.308	0.278	0.255	0.236	0.221	0.208
	40	0.422	0.351	0.303	0.269	0.243	0.222	0.206	0.193	0.181
	50	0.360	0.300	0.259	0.230	0.207	0.190	0.176	0.165	0.155
	60	0.299	0.249	0.215	0.190	0.172	0.158	0.146	0.136	0.129
	70	0.237	0.198	0.171	0.151	0.137	0.125	0.116	0.108	0.102
9.5	-10	0.574	0.479	0.414	0.367	0.332	0.304	0.282	0.264	0.248
	0	0.525	0.438	0.379	0.336	0.304	0.279	0.258	0.241	0.227
	10	0.477	0.398	0.344	0.305	0.276	0.253	0.234	0.219	0.206
	20	0.428	0.358	0.309	0.274	0.248	0.227	0.211	0.197	0.185
	30	0.380	0.317	0.274	0.243	0.220	0.202	0.187	0.175	0.165
	40	0.332	0.277	0.239	0.212	0.192	0.176	0.163	0.152	0.144
	50	0.283	0.236	0.205	0.181	0.164	0.150	0.139	0.130	0.123
	60	0.235	0.196	0.170	0.151	0.136	0.125	0.116	0.108	0.102
	70	0.187	0.156	0.135	0.120	0.108	0.099	0.092	0.086	0.081

(Note) The table above is applicable under the following condition:

- Use of thermal paper "TF50KS-E2D"
- Vp and GND wiring resistance : $r_c = 0$
- The number of simultaneously activated dots : $N_{SA} = 45$

3.5.8 Temperature Characteristics of Thermistor

Calculate the resistance of the thermistor (R_X) at the operating temperature T_X (°C) using the following equation (9). Variation of resistance by temperature is shown in Figure 3-10 and Table 3-14.

Equation (9):

$$R_x = R_{25} \times \mathrm{ESP}\left\{B \times \left(\frac{1}{273 + T_x} - \frac{1}{298}\right)\right\}$$

R_X	:	Resistance at T_X °C (Ω)	
R_{25}	:	Resistance at 25 °C	30.00 kΩ ± 5%
B	:	B constant	3950 K ± 3%
T_X	:	Temperature detected by thermistor (°C)	
EXP (A)	:	The "A" th power of natural logarithm e (2.71828)	

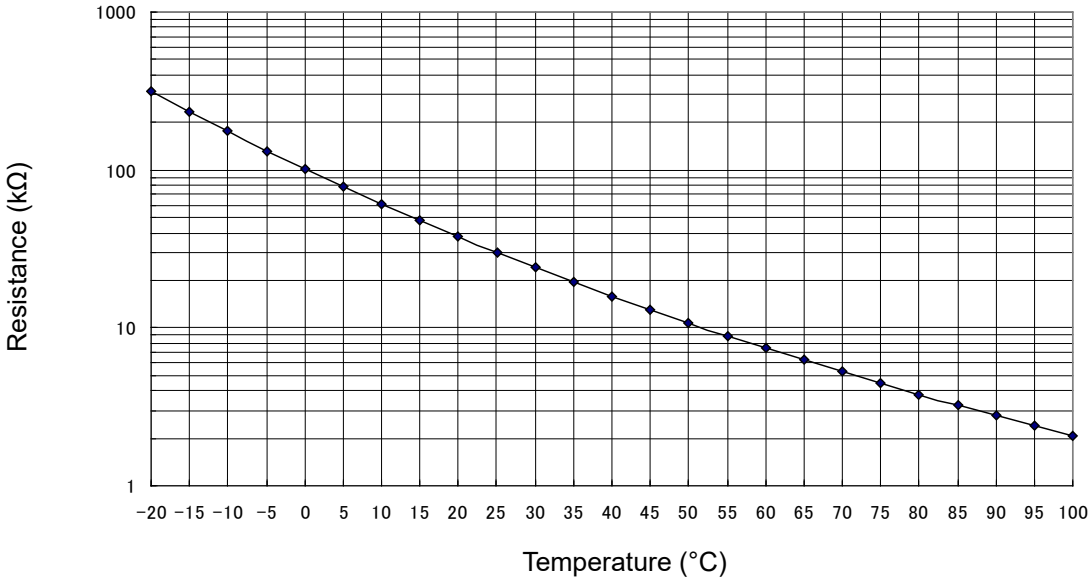


Figure 3-10 Temperature Characteristics of Thermistor

Table 3-14 Temperature Characteristics of Thermistor

Temperature (°C)	Thermistor Resistance (kΩ)
-10	175.07
-5	132.29
0	100.99
5	77.85
10	60.57
15	47.53
20	37.61
25	30.00
30	24.11
35	19.51
40	15.89
45	13.03
50	10.75
55	8.92
60	7.45
65	6.25
70	5.27
75	4.47
80	3.80
85	3.25
90	2.79
95	2.41
100	2.09

3.5.9 Detecting Abnormal Temperature of Thermal Head

To protect the thermal head and to ensure personal safety, abnormal temperature of the thermal head must be detected by both hardware and software as follows:

(1) Detecting Abnormal Temperatures by Software

Design the software so that energization to the thermal head can be stopped until the temperature becomes 75°C or lower (thermistor resistance $R_X \geq 4.47$ kΩ) when the thermistor (TH) of the thermal head detects 85°C or higher ($R_X \leq 3.25$ kΩ).

(2) Detecting Abnormal Temperatures by Hardware

When such as the control unit (CPU) runaway occurs, and energization to the thermal head is continued, the temperature detection of the software may not function properly and result in overheating of the thermal head. Overheating of the thermal head not only may damage the thermal head but also may cause smoke, fire and burn injuries. Always use hardware together with software for detecting abnormal temperatures to ensure personal safety. (In the case of the control unit runaway, the thermal head may be damaged even when the abnormality detection is performed by hardware.)

In hardware, detect the following conditions by using a detection circuit such as a window comparator circuit.

(a) Overheating of the thermal head
(approximately 100°C or higher ($R_X \leq 2.09$ kΩ))

(b) Faulty thermistor connection (the thermistor may be open or short-circuited).

In case of abnormality, turn off immediately the power supply. Reactivate the thermal head after returning from the abnormal state.

3.6 OUT-OF-PAPER SENSOR

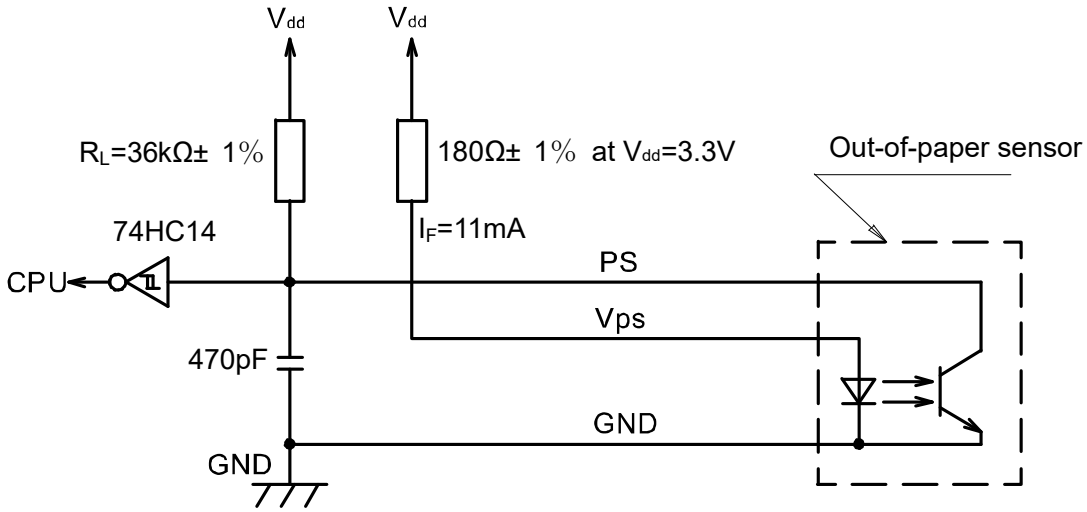
The printer has a built-in out-of-paper sensor (reflection type photo interrupter) to detect whether paper is present or not. An external circuit should be designed so that it can detect output from the out-of-paper sensor and does not activate the thermal head nor drive the motor when there is no paper. Activating the thermal head with no paper state may cause damage to the thermal head or platen roller or shorten the life of the thermal head significantly. If the motor is driven with no paper, a load is put on the paper feed drive system and the life of the printer may be shortened significantly.

Table 3-15 shows the model of the out-of-paper sensor used for this printer.

Table 3-15 Out-of-paper Sensor

Item	Specification
Type	ITR1204SR10A/TR(KCR-1)
Manufacturer	EVERLIGHT ELECTRONICS CO.,LTD

Figure 3-11 shows an example of the external circuit of the out-of-paper sensor.



*: The PS signal is "High" when there is no paper.

Figure 3-11 Example External Circuit of Out-of-paper Sensor

3.6.1 Precautions for Out-of-Paper Sensor

- The out-of-paper sensor may generate instantaneous abnormal signal. Design the firmware not to produce improper operating due to the abnormal signal. (Recommended frequency: detecting twice every 10 ms.)

CHAPTER 4
CONNECTING TERMINALS

4.1 RECOMMENDED CONNECTOR FOR EXTERNAL CIRCUITS

Use the recommended connectors listed in Table 4-1 to connect the connecting terminals firmly to the external circuits.

Table 4-1 Recommended Connectors

Number of Terminals	Recommended Connectors
24	MOLEX INC: 0545502471 (Right angle type : Top contact, Gold plated)

4.2 CONNECTING TERMINALS

Figure 4-1 shows the terminal configuration of the connecting terminals and Table 4-2 shows terminal assignments of the connecting terminals.

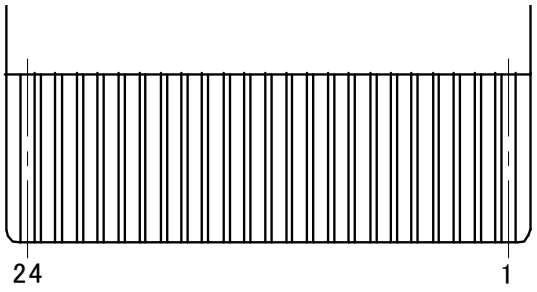


Figure 4-1 Connecting Terminals

Table 4-2 Terminal Assignments of Connecting Terminal

Terminal No.	Terminal Name	Function
1	PS	Output of the out-of-paper sensor (Photo-transistor collector)
2	G_{PS}	GND of the out-of-paper sensor (LED cathode, photo-transistor emitter)
3	V_{PS}	Power supply of the out-of-paper sensor (LED anode)
4	V_p	Thermal head drive power supply
5	V_p	Thermal head drive power supply
6	V_p	Thermal head drive power supply
7	DI	Print data input (serial input)
8	CLK	Synchronizing for print data transfer
9	GND	GND
10	GND	GND
11	GND	GND
12	V_{dd}	Logic power supply
13	DST	Thermal head print activation instruction
14	TH	Thermistor
15	GND	GND
16	GND	GND
17	GND	GND
18	/LAT	Print data latch (memory storage)
19	V_p	Thermal head drive power supply
20	V_p	Thermal head drive power supply
21	A	Motor drive
22	/A	Motor drive
23	B	Motor drive
24	/B	Motor drive

CHAPTER 5
PRINT DRIVE METHOD

5.1 DRIVE METHOD OF MOTOR AND THERMAL HEAD

In order to print, the motor and the thermal head should be driven at the same time.

The printer uses a half dot pitch thermal head with which the paper feed pitch of the heat element is 1/2 dot, so 2 half dots form 1 dot. Also, the motor feeds paper for 1 dot-line in 4 steps. Therefore, it is necessary to feed paper for 4 steps and activate the thermal head once every 2 steps to form 1 dot-line.

Figure 5-1 shows a divided print timing chart example.

The 1st dot-line shows the case of 3 divisions and the 2nd dot-line shows the case of batch printing.

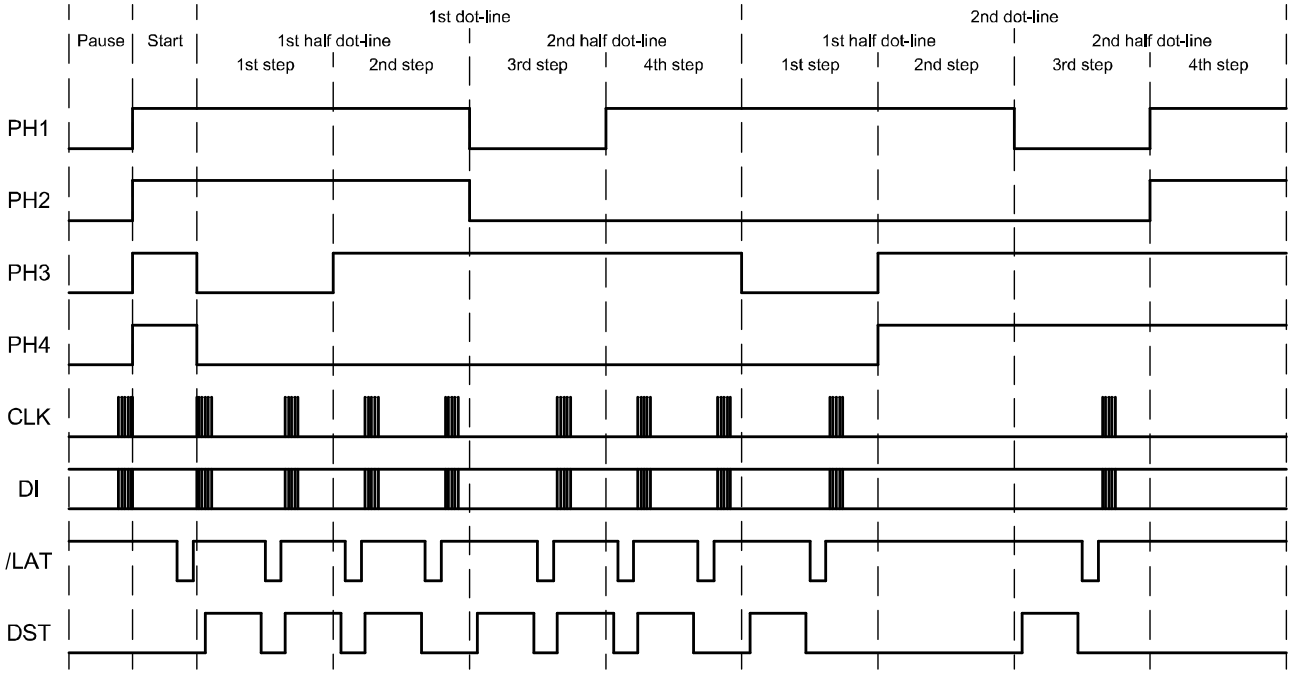


Figure 5-1 Divided Print Timing Chart Example

The drive method with divided printing is described below (See Figure 5-1):

(1) Pause State

Do not excite the motor. Always set /DST of the thermal head to "Low".

(2) Start Step

Excite the phase that was output just before the motor was stopped.

(3) 1st Dot-line

Activate the thermal head once every 2 steps of the motor drive signal. Configure 1 dot-line by 4 steps of the motor drive signal.

At the 1st step of the motor drive signal, start activation of the thermal head by synchronized the DST signal and print the 1st half dot-line at the 1st dot-line by DST.

After 1st step of the motor drive signal is completed, input the 2nd step of the motor drive signal. (It is not necessary to synchronize the activation of the thermal head.)

After printing the 1st half dot-line in the 1st dot-line, start activation of the thermal head by synchronized the DST signal to the 3rd step of the motor drive signal and then print the 2nd half dot-line in the 1st dot-line by DST.

After 3rd step of the motor drive signal is completed, input the 4th step of the motor drive signal. (It is not necessary to synchronize the activation of the thermal head.)

Transfer the data to be printed to SHIFT REGISTER of the thermal head prior to the input of the DST and latch to LATCH REGISTER of the thermal head by the input of the /LAT signal.

(4) Procedures That Follows the 2nd Dot-line

Proceed with the step in the same way as the 1st dot-line, and repeat the motor driving and thermal head activation.

5.2 DIVISION DRIVE METHOD OF THERMAL HEAD

The printer uses the dynamic division method. The dynamic division method is the method that determines the logical block not to exceed the maximum number of set drive dots when printing 1 dot-line. The determination of the logical block is performed for every 1 dot-line print. The maximum current consumption can be always controlled within a constant value with this method.

The thermal head in the printer can be driven by 1 DST signal. Drive the thermal head with the number of simultaneously activated dots of 45 dots or less as a logical block.

5.3 PRECAUTIONS FOR PRINT DRIVE

- When printing all physical blocks with a batch process, secure a pause time of 0.5 ms or more after activating the thermal head and before activating the same heat element.
- The number of the maximum thermal head division in a half dot-line should be 9 or lower to maintain print quality. The number of the simultaneously activated dots should be 45 dots or less.

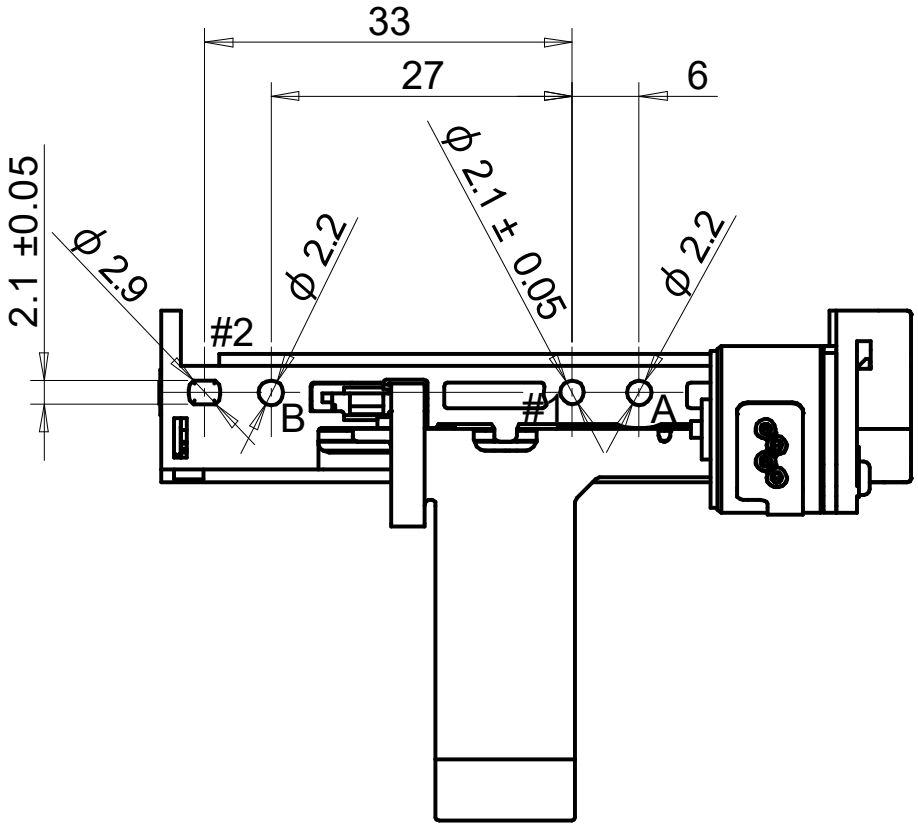
CHAPTER 6
OUTER CASE DESIGN GUIDE

6.1 FIXING PRINTER MAIN BODY

6.1.1 How to Mount Printer Main Body

Figure 6-1 shows the mounting dimensions of the printer main body.

- #1 and #2 are positioning holes. Design bosses on the outer case for the positioning holes #1 and #2. The height of the bosses on the outer case should be 1.5 mm (max.) or less.
- Screw up holes A and B.



Unit : mm
General tolerance : ±0.1

Figure 6-1 Mounting Dimensions of Printer Main Body

6.1.2 Recommended Screws

- JIS B1111 M2.0 cross-recessed pan head machine screw

6.1.3 Precautions for Fixing Printer Main Body

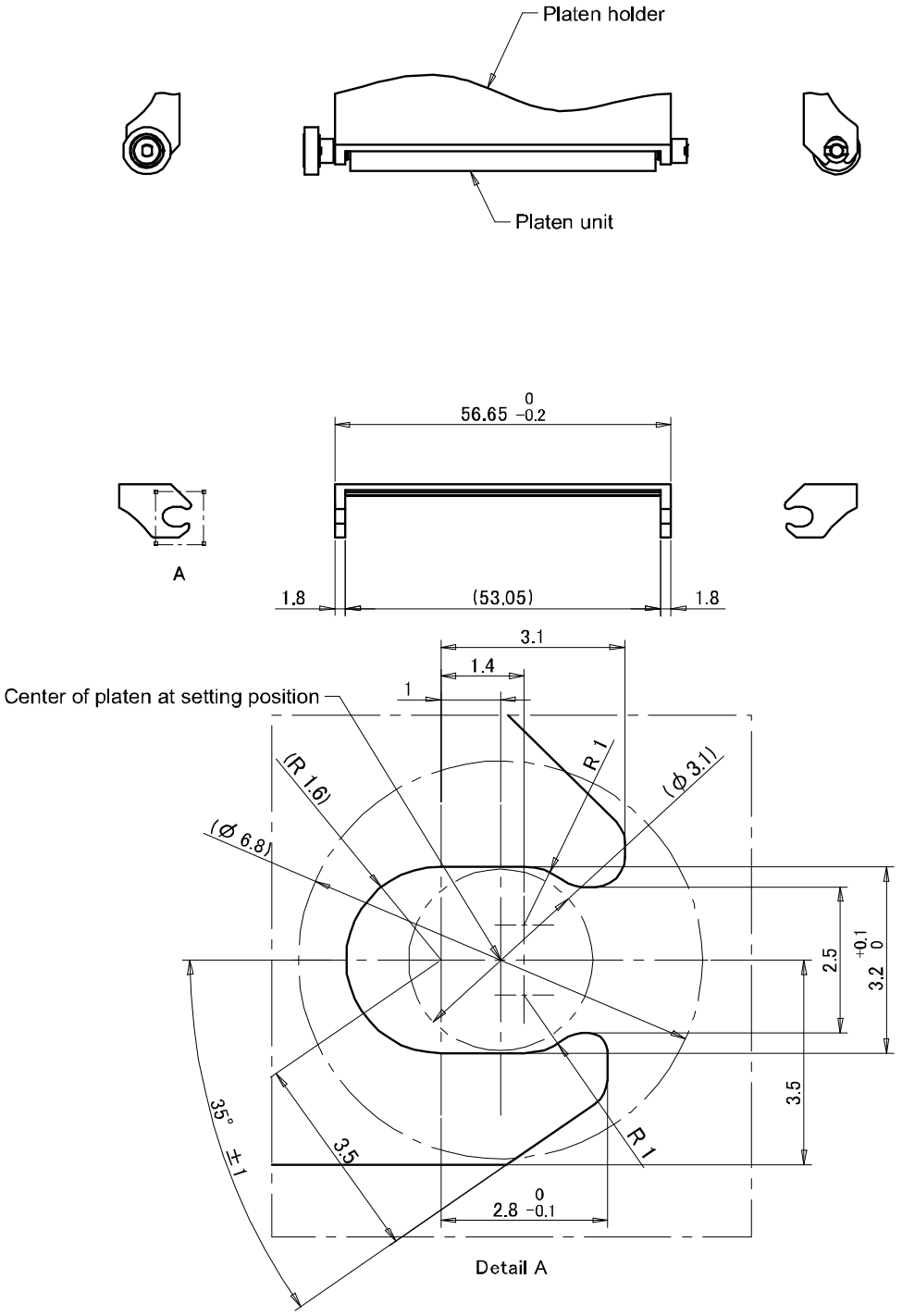
- Fix the printer main body with care so that it will be not applied an excessive force, deformation, or torsion. The deformed or twisted printer main body can cause print deterioration, paper skew, paper jam, or noise.
- Install the printer main body on a flat surface and use it avoiding places where vibrations are caused.
- Connect the Frame Ground (FG) to Signal Ground (GND) through approximately 1 MΩ resistance, and make the same electric potential.
  See "6.3 CONNECT TO FRAME GROUND (FG)" for details about how to connect the Frame Ground (FG).
- Pay attention not to damage on the FPC when fixing the printer main body.

6.2 FIXING PLATEN UNIT

For mounting the platen unit, design the platen holder on the outer case.

6.2.1 Recommended Shape of Platen Holder

- Figure 6-2 shows a recommended shape of the platen holder.

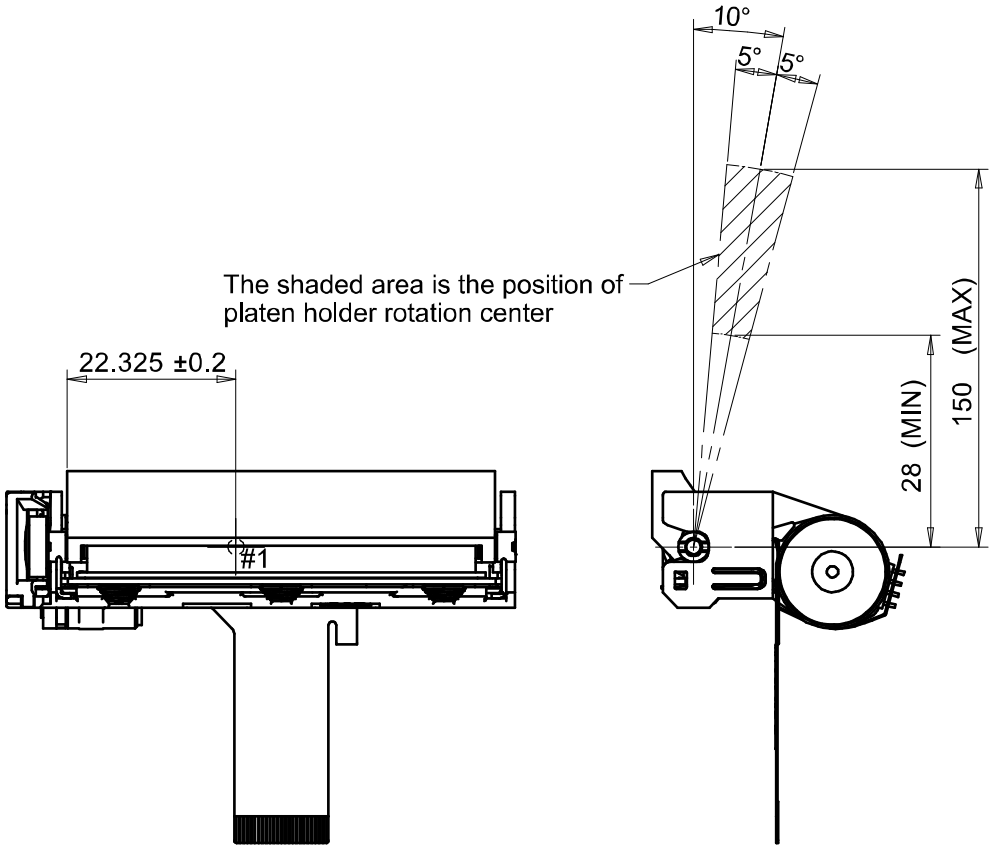


Unit : mm
General tolerance for dimensions : ±0.1
General tolerance for angles : ±1°

Figure 6-2 Recommended Shape of Platen Holder

6.2.2 Mounting Dimensions of Platen Holder

Figure 6-3 shows the range of the rotation center of the platen holder, which is critical for opening and closing the platen unit.



Unit : mm
General tolerance for dimensions : ±0.1
General tolerance for angles : ±1°

Figure 6-3 Dimensions for Positioning and Fixing Platen Holder

6.2.3 Precautions for Mounting Platen Holder

- Design the platen holder not to contact with the thermal head.
- Setting the mounting part of the printer and rotation center part of the platen holder to the same member, improves accuracy of opening and closing the platen unit.
- It is recommended that the platen holder is guided by part of the outer case, when the platen unit is set into the printer main body.
- Design the rotation center of the rotation holder opening and closing mechanism as shown in Figure 6-3 based on at 10° as a central line horizontally from the surface of the printer body mounting, above and below between 5° and between 28 mm to 150 mm from the platen center. The shaded area in the figure is the range of the center of the platen holder rotation.
- Parallelism between the positioning holes #1-#2 and the rotation center axis of platen holder should be 0.2 or less.
- Design the platen holder to be strong enough to stand against stress applied during opening and closing the platen unit.
- Fix the printer main body with care so that it will not be applied an excessive force, deformation, or torsion.
- When the mounting the position of the platen holder and the position of the platen unit are not proper, poor-fitting between the printer main body and platen unit may occur, resulting in print failure or paper jam.
- Do not use metal material for the platen holder. When using the metal material, the platen unit cannot be mounted.
- Design the outer case and the platen holder to push the center of the platen unit. When the edge of the platen unit is pushed, only one side of the platen unit is locked, it may cause defects such as print failure or paper jam. In order to let users to push the center area, any mark indication is effective. Verify the performance with your actual device in advance.
- Do not apply external force continuously toward the direction the platen unit is opened by such as a spring. Otherwise, it may cause slipping gears, negative influence on printing quality, and printing troubles during operation.

6.3 CONNECT TO FRAME GROUND (FG)

To prevent the thermal head from being damaged by static electricity, it is recommended that the printer mechanism is connected to the frame ground (FG).

6.3.1 How to Connect Printer to Frame Ground (FG)

- Connect the connection member of the frame ground (FG) to the part indicated by the arrow as shown in Figure 6-4.
- Design the connecting parts with tension like coil spring or plate spring to reduce the contact resistance.
- Design the connection member so as to apply contact pressure in the direction of the arrow as shown in Figure 6-4, and horizontally to the mounting surface of the printer.
- Use beryllium copper or other material with superior electrical conductivity for the connecting members.
- The connecting force should be 1N or lower.
- All Frame Ground (FG) should be the same electrical potentials.
- Connect the signal ground (GND) to the frame ground (FG) through approximately 1 MΩ resistor to make the same electric potentials.

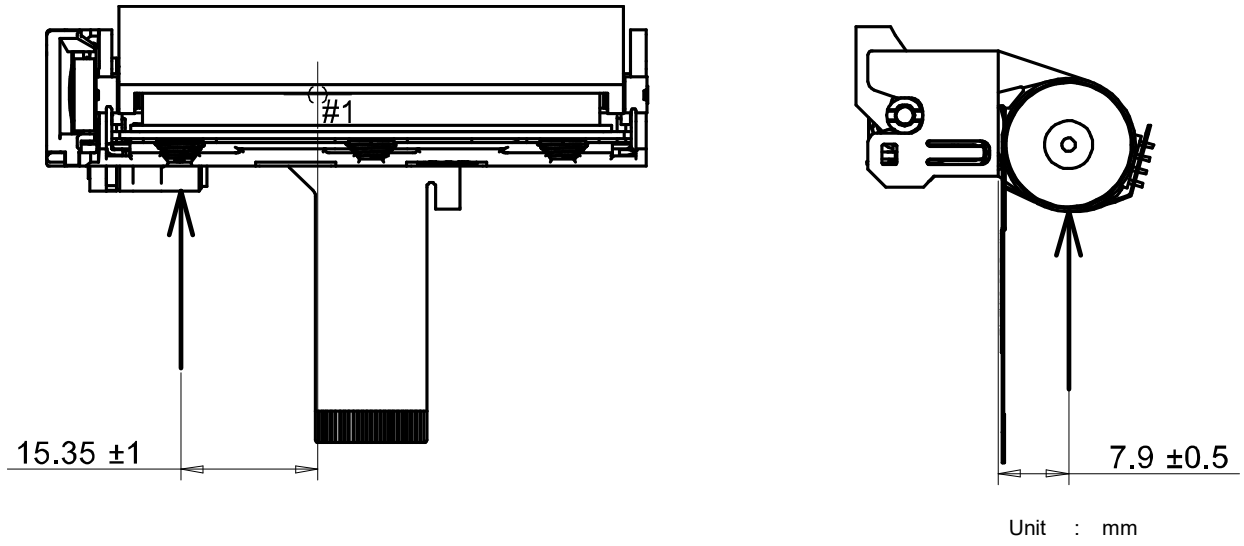


Figure 6-4 Frame Ground (FG) Connector Part

6.4 LAYOUT OF PRINTER MECHANISM AND THERMAL PAPER

The printer mechanism can be laid out as shown below.

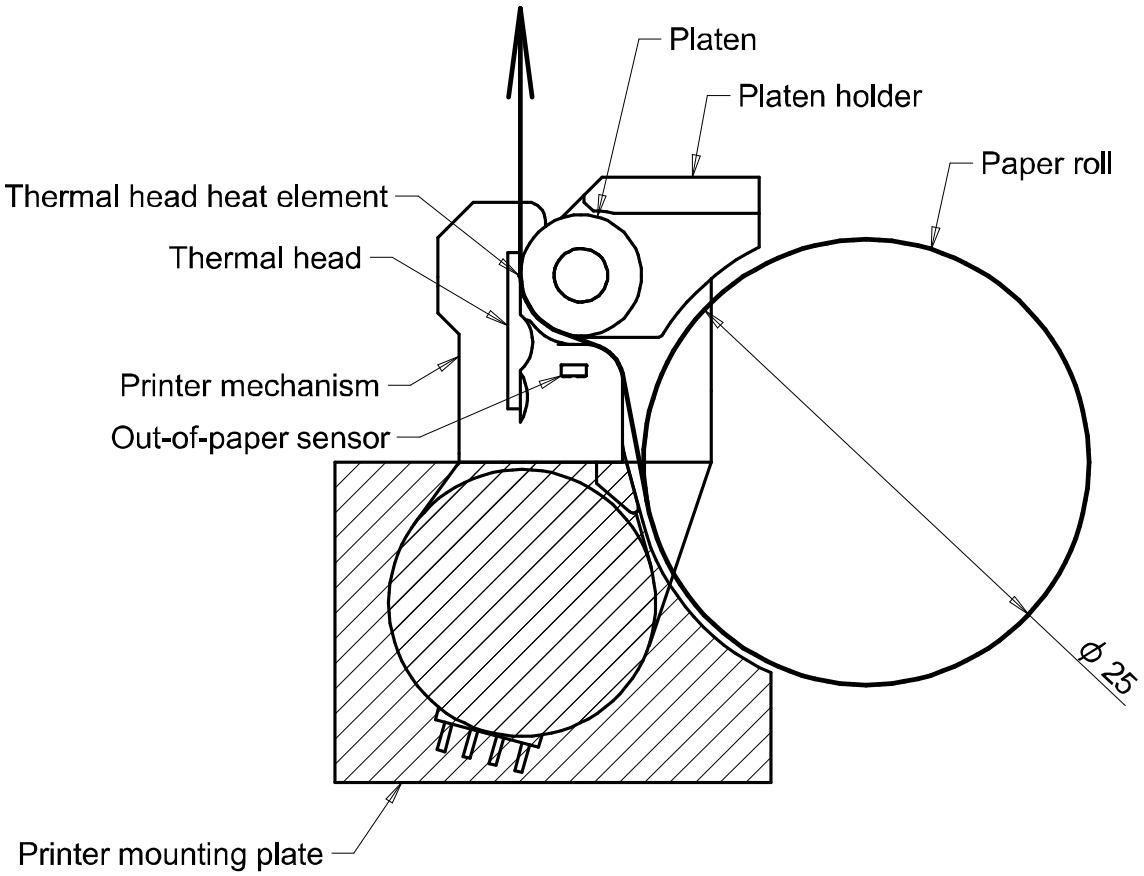


Figure 6-5 Layout of Printer Mechanism and Thermal Paper

(Note) The length between the out-of-paper sensor and the heat element is approximately 5.0 mm.

6.5 MOUNTING POSITION OF PAPER HOLDER

When designing the layout of the paper holder, note the followings. The recommended shape of the paper holder is shown in Figure 6-6.

- When using roll type thermal paper, design the paper holder and the paper guide to make the thermal paper to be straight to the paper inlet port without any horizontal shifting and to make the center axis of the paper roll to be parallel to the printer.

- The paper feed load to the printer should be 0.49 N or less. However, the printing failure may occur in the following case even when the paper supply load is 0.49 N or less. Design the paper holder with paying attention to the following points, and verify the performance with your actual device in advance.

  e.g.)
  When the paper roll wobbles in the paper holder during printing.
  When the paper tension or loosen fluctuates significantly during printing.

  In addition, do not use following types of paper roll:

  - There is winding looseness.
  - The shape is deformed.
  - The core is sticking out.
  - The width is out of spec.

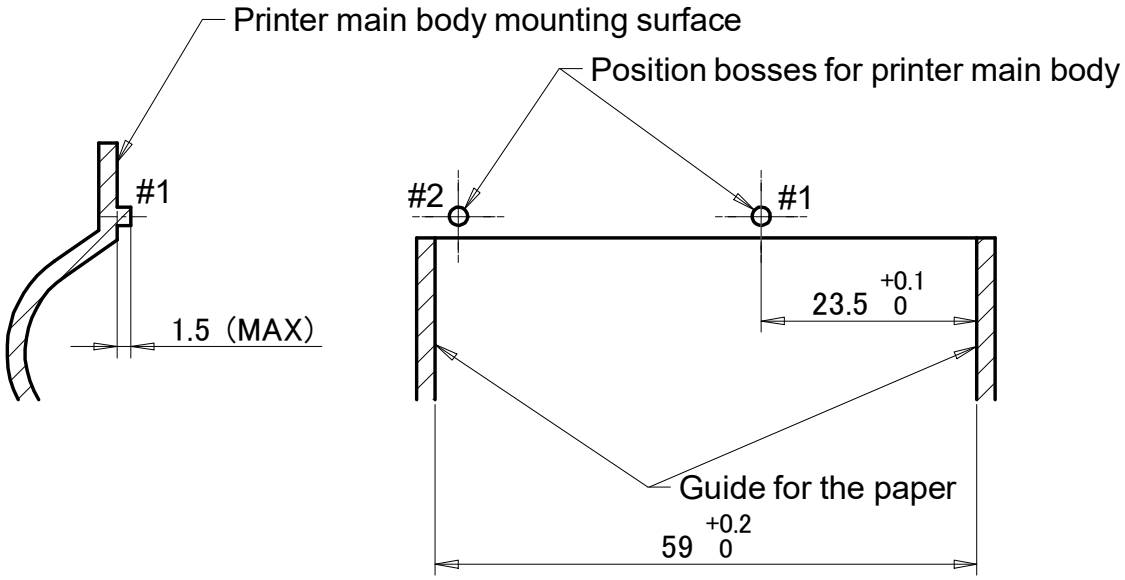


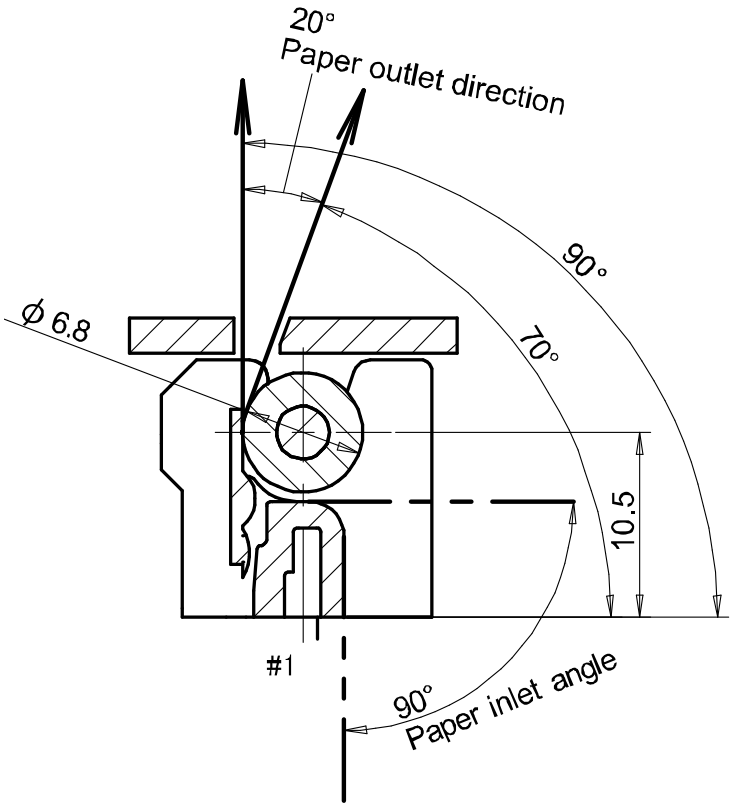
Figure 6-6 Recommended Paper Holder Dimensions

6.6 DESIGN PAPER PATH

6.6.1 Shape of Paper Outlet

When designing the shape of the paper outlet, pay attention to the followings.

- Design the shape of the paper outlet not to apply any stress to the thermal paper to be ejected.
- Design the shape of the paper outlet of the outer case to make the paper outlet angle be 70° to 90° as shown in Figure 6-7. However, design the paper outlet to be able to freely outlet thermal paper without changing the direction of the paper outlet. Especially right after ejecting paper from the printer mechanism, do not change the paper outlet direction. If the paper outlet direction is to be changed, verify the performance with your actual device in advance.



Unit : mm
General tolerance for dimensions : ±0.1

Figure 6-7 Paper Path

6.6.2 Mounting Paper Cutter

- Design the paper cutter so that its cutting edge will not contact the platen unit that opens and closes.
- Use a well-cut cutter so that the thermal paper can be cut with less force than paper holding force.
- If the paper cutter is designed as on the left side shown in Figure 6-8 the thermal paper after cutting may be caught on the cutting edge and caught in the inside of the cutter. As the cutting edge shown on the right in the figure, design the shape of the cutter so that it can guide the thermal paper after cutting.

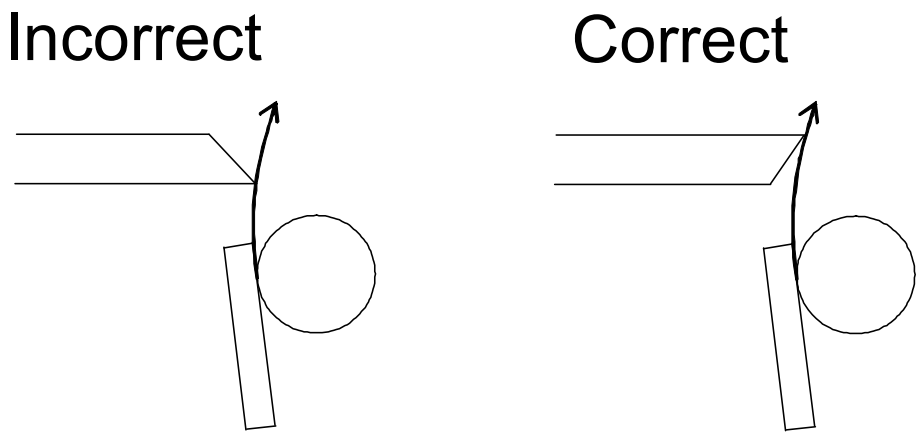


Figure 6-8 Cutting Edge Shape of Paper Cutter

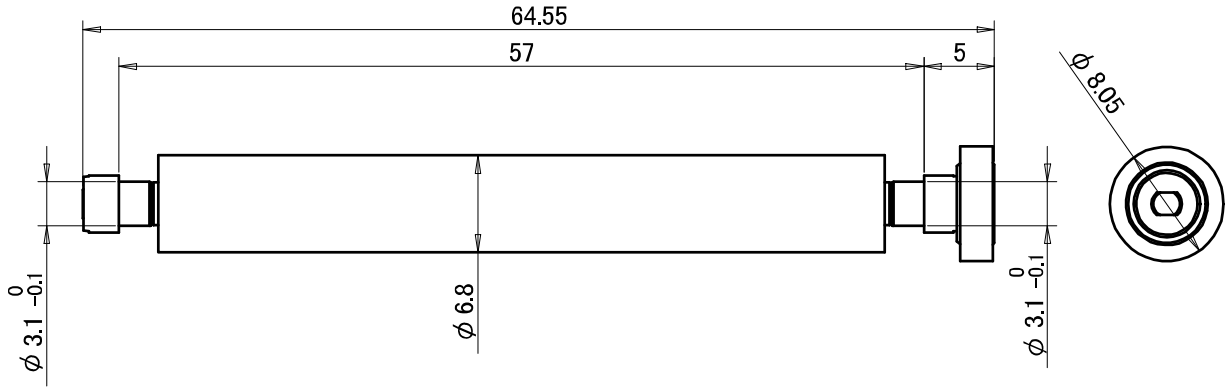
6.7 PRECAUTIONS FOR DESIGNING OUTER CASE

- The thermal paper with a small winding diameter may cause a paper jam in the printer main body or between the printer and the outer case. When such thermal paper is necessary to be used, verify the performance with your actual device in advance.
- Design the outer case not apply any load to the printer main body and the platen unit. The load is applied, it may cause print defects, paper jams, and/or damage to the printer. To prevent this, secure at least 1.0 mm space between the outer case and the printer main body or platen unit.
- Temperature of the thermal head and its peripherals rises very high during and immediately after printing. Be sure to design the outer case to prevent users from burn injuries by directly touching them and call attention of users by giving sufficient warning with a label or other. Sufficiently warn users for cleaning the thermal head after the thermal head has cooled down. In order to effectively release heat, provide space between the thermal head and the outer case.
- Temperature of the motor and its peripherals rises very high during and immediately after driving. Be sure to design the outer case to prevent users from burn injuries by directly touching them and call attention of users by giving sufficient warning with a label or other. In order to effectively release heat, provide space between the thermal head and the outer case.

CHAPTER 7
EXTERNAL DIMENSIONS

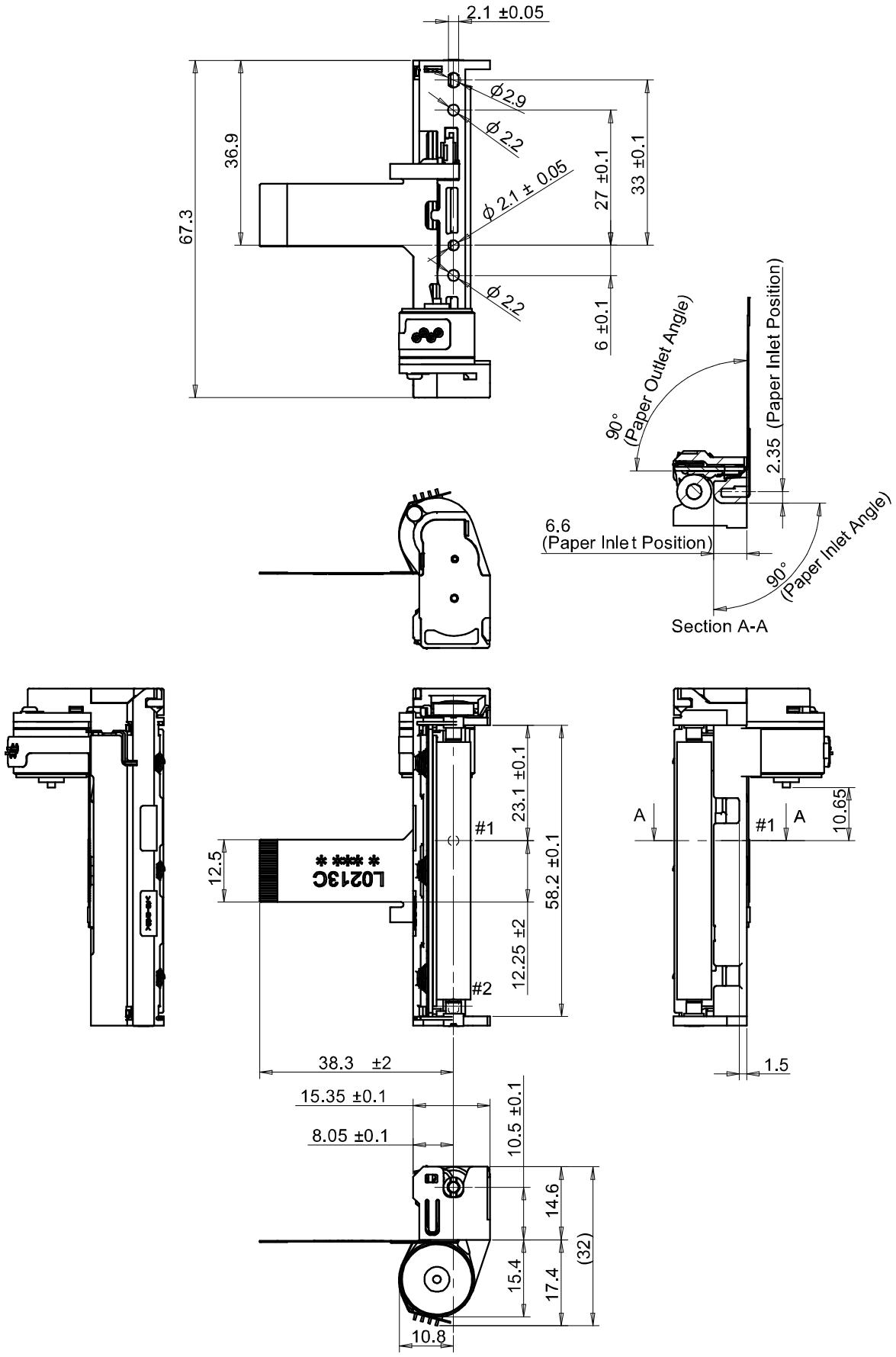
7.1 EXTERNAL DIMENSIONS

Figure 7-1 shows external dimensions of the platen unit. Figure 7-2 shows external dimensions of the printer mechanism.



Unit : mm
General tolerance for dimensions : ± 0.5

Figure 7-1 Platen Unit External Dimensions

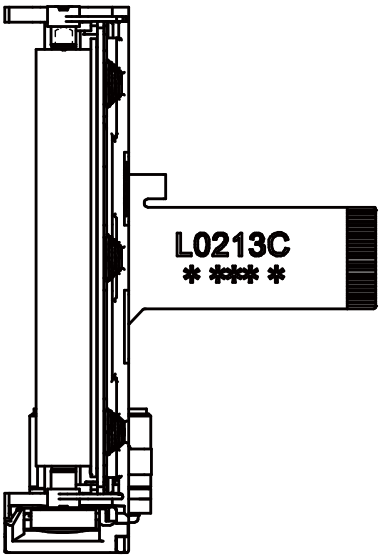


Unit : mm
General tolerance for dimensions : ±0.5

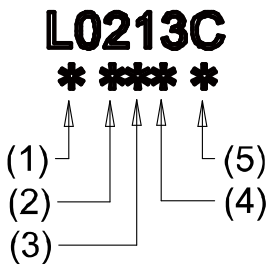
Figure 7-2 Printer Mechanism External Dimensions

7.2 MODEL INDICATION

7.2.1 Model Number Locations



7.2.2 Code Description



(1) Internal code

(2) Manufacturing year code (last 1 digit)

Code	A	B	C	D	E	F	G	H	J	K
Year	1	2	3	4	5	6	7	8	9	0

(3) Manufacturing month code

Code	A	B	C	D	E	F	G	H	J	K	L	M
Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec

(4) Manufacturing week code

Code	1	2	3	4	5
Week	1 to 6	7 to 12	13 to 18	19 to 24	25 to 31

(5) Manufacturing date code

<e.g.> June 30, 2025 → *EF5*

CHAPTER 8
HANDLING METHOD

8.1 PRECAUTIONS FOR INSTALLING/UNINSTALLING THERMAL PAPER

- Install the thermal paper with the paten unit opened. The printer does not allow the installation of the thermal paper by auto-loading.
- Set the thermal paper straightly to make the paper edge 5 cm or more out from the upper surface of the printer mechanism.
- When setting the platen unit, the tips of the platen gear and those of the reduction gear may contact and the platen cannot be installed. In such a case, open the platen unit, and close it again.
- When the thermal paper is skewed, feed the thermal paper until the thermal paper becomes straight or install the thermal paper again.
- When removing the thermal paper due to such as a paper jam, remove the thermal paper with the platen unit opened. Do not forcibly pull on the thermal paper, it may cause damages.

8.2 PRECAUTIONS FOR CLEANING THERMAL HEAD

- Do not clean the thermal head immediately after printing because the temperature of the thermal head and its peripherals rises very high during and immediately after printing.
- Clean the thermal head with the platen unit opened.
- Clean the thermal head with a cotton swab soaked with ethyl alcohol or isotropy alcohol and wipe off the dirt (shown in Figure 8-1).
- Do not use anything which may damage the heat element such as sandpaper, or a cutter knife when cleaning it.
- Set the platen unit after the alcohol has dried completely.

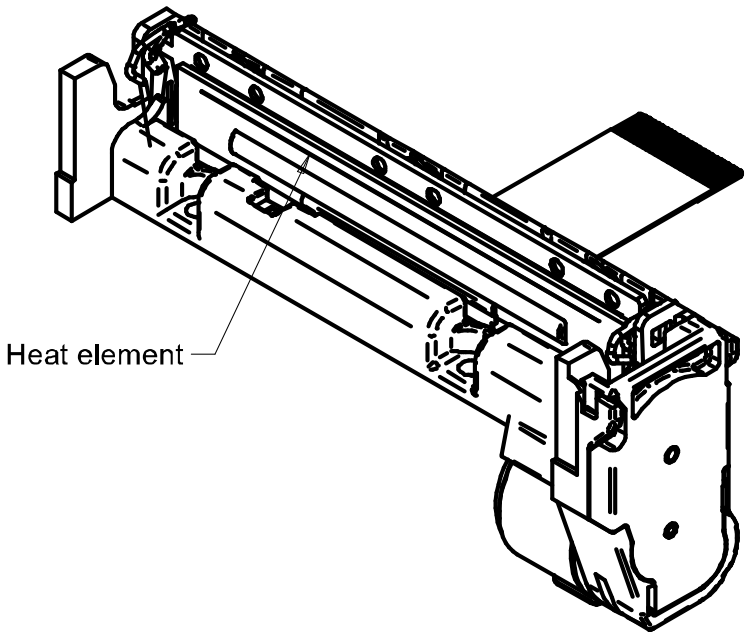


Figure 8-1 Cleaning Position of Thermal Head